

OPEN ACCESS

Is Silicon Replaceable? A Physical, Chemical, and Electrochemical Analysis of Different Commercial Lithium-Ion Battery Cells

To cite this article: Nicolas Stapf *et al* 2025 *J. Electrochem. Soc.* **172** 060513

View the [article online](#) for updates and enhancements.

You may also like

- [Zr and Zr-Cr Commercial Conversion Coatings Deposited on 3003 Aluminium Alloy Foil](#)
Maja Mjuidrica Kim, Barbara Kapun and Ingrid Milošev
- [Degradation Mode Analysis for Lithium-Ion Cells with Silicon-Dominant Anodes Using Reference Electrodes](#)
S. Friedrich, F. Dengler, M. Bock et al.
- [Correlating the Improved H₂/Air Performance of Mesoporous Graphitic Spheres with their Tailored Pore Structure](#)
Roberta K. F. Della Bella, Alexander P. Gunnarson, Marc-Georg Willinger et al.

Your Lab in a Box!

The PAT-Tester-i-16 Multi-Channel Potentiostat for Battery Material Testing!

- ✓ **All-in-One Solution with Integrated Temperature Chamber (+10 to +80 °C)!**
No additional devices are required to measure at a stable ambient temperature.
- ✓ **Fully Featured Multi-Channel Potentiostat / Galvanostat / EIS!**
Up to 16 independent battery test channels, no multiplexing.
- ✓ **Ideally Suited for High-Precision Coulometry!**
Measure with excellent accuracy and signal-to-noise ratio.
- ✓ **Small Footprint, Easy to Setup and Operate!**
Cableless connection of 3-electrode battery test cells. Powerful EL-Software included.



EL-CELL®
electrochemical test equipment

Learn more on our product website:



Download the data sheet (PDF):



Or contact us directly:

+49 40 79012-734

sales@el-cell.com

www.el-cell.com



Is Silicon Replaceable? A Physical, Chemical, and Electrochemical Analysis of Different Commercial Lithium-Ion Battery Cells

Nicolas Stapf,^z  Nicolas Schneider,  Meichen Zhan,  Christopher Mehlich, 
Kai Peter Birke,  and Alexander Fill 

University of Stuttgart, Institute for Photovoltaics, Chair for Electrical Energy Storage Systems 70569 Stuttgart, Germany

The addition of silicon to graphite electrodes generally enhances energy density but shortens cycle life due to the volume changes during (de-)lithiation. To balance these effects, commercial cells incorporate small silicon amounts. This study investigates the physical, chemical, and electrochemical properties of three cylindrical lithium-ion cells—the LG INR21700-M50, M50T, and M50LT—on particle, electrode, and cell level. After cell teardown and sample preparation, i.e. energy-dispersive X-ray spectroscopy (EDS) and electrochemical investigations reveal differences in the cells' electrodes: While the M50 and M50T employ graphite/silicon composite anodes, the M50LT solely contains graphite as an active material. Despite the absence of silicon, the M50LT achieves a comparable cell-level capacity, attributed to increased nickel content in the positive electrode. This raises the question of silicon's replaceability, addressed throughout the study and summarized in the conclusion. The study also determines the silicon content in the composite electrode of the LG M50T to 2.43% using five complementary, cross-validated methods. Additional comparisons include the open circuit potentials (OCP), particle size distributions, and solid diffusion coefficients. This study characterizes the three cells to support the interpretation of related research and provide reference data for future studies applicable also for electrochemical model parametrization.

© 2025 The Author(s). Published on behalf of The Electrochemical Society by IOP Publishing Limited. This is an open access article distributed under the terms of the Creative Commons Attribution 4.0 License (CC BY, <https://creativecommons.org/licenses/by/4.0/>), which permits unrestricted reuse of the work in any medium, provided the original work is properly cited. [DOI: 10.1149/1945-7111/add112]



Manuscript submitted January 14, 2025; revised manuscript received April 9, 2025. Published June 11, 2025.

Supplementary material for this article is available [online](#)

Motivation.—For mobile electrified transportation such as electric vehicles or aviation, the energy density of the battery is crucial. Increasing the energy density of battery packs can either be done by reducing the weight of the subcomponents of a battery system, or by increasing the storable energy per weight inside the single battery cells. The latter is achievable by increasing the voltage of the battery or by using active materials with a higher specific capacity. Silicon or silicon oxide (SiO_x , $0 < x \leq 2$) is an attractive active material which includes both a low voltage level for the negative electrode operation resulting in a high full-cell voltage and a high energy density of 3579 mAh g^{-1} – 4200 mAh g^{-1} depending on which lithiation state is considered.^{1–3} The theoretical energy density is about ten times higher than that of state of the art graphite electrodes (372 mAh g^{-1}).⁴ Nonetheless, silicon particles undergo high volume changes of about 300% while being (de-)lithiated which leads to mechanical stress in the electrode particles.⁵ Consequently, the particles and the surrounding solid electrolyte interphase (SEI) tend to crack and needs to be reformed again on the newly exposed surfaces. These effects reduce the cycleable capacity and hence the life time of silicon-containing electrodes.^{6–8} To compensate this dilemma, silicon oxide can be used instead of pure silicon since it only expands by about 118% while being (de-)lithiated.⁵ Irreversible reaction products during the first cycles act as a buffer and compensate the volume expansion, however, the theoretical capacity is reduced to about 1700 – 2400 mAh g^{-1} .^{9,10} Another option to increase cycle life is to completely omit silicon or silicon oxide. However, the lower specific capacity of the battery cell must then be accepted.¹⁰ Silicon and silicon oxide were already inserted in commercialized battery cells such as the LG Chem INR18650-MJ1, the Sony US18650-VTC6, the Samsung SDI INR21700-50E, the Samsung SDI INR21700-50S, or the Samsung SDI INR18650-30Q.^{11–14} The commercially available cylindrical battery cell LG Chem INR21700-M50 also contains silicon oxide as characterized by Chen et al.¹⁵ and stands out due to its high cell capacity of 5 Ah in the 21700 format. According to the datasheet, the

LG M50 was introduced in 2016 and upgraded to the LG M50T in 2018 and finally to the LG M50LT in 2021 which is currently the latest version.

The LG M50 and its subvariants were used in different publications for simulation and experimental purposes. For instance, the behavior of the LG M50T was simulated in Ref.^{6,16–20}, using the pseudo-two-dimensional (p2D) approach from Doyle-Fuller-Newman.²¹ The degradation was investigated in Ref.^{22,23} by varying the ambient temperature, the C-rates and the depth of discharge (DoD). The LG M50LT was also taken for design analysis, aging determinations, thermal runaway studies and state estimations.^{24–28} Often, the parameters determined in the characterization from Chen et al.¹⁵ for the LG M50 were used to build up simulation models or conduct experiments assuming that the LG M50T and LG M50LT have the same properties as the previously designed LG M50.

Therefore, this study analyzes the physical, chemical and electrochemical behavior of the different subvariants of the LG M50—the M50T and M50LT—by various measurement techniques and compares the results with the characterization from Chen et al. for the LG M50.¹⁵ Based on the comparison of the individual subvariants of the M50, it is discussed whether the addition of silicon or silicon oxide in lithium-ion cells is replaceable. The study also aims to provide relevant electrode properties to determine the parameters for p2D models for the LG M50T and LT using the approach from Doyle-Fuller-Newman.²¹

Relevant literature.—To contextualize the present work, several studies have examined commercial battery cells, also with a silicon-containing negative electrode, with respect to their internal properties. Chen et al. (2020)¹⁵ provided an extensive physical, chemical and electrochemical characterization of the LG M50 cell, including energy-dispersive X-ray spectroscopy (EDS)-based visualization of graphite and silicon oxide in the negative electrode and particle size examination via scanning electron microscope (SEM) images. Their aim was to determine the required parameters to electrochemically model the battery cell. Their work forms a crucial basis for subsequent modeling studies by offering cell-specific input parameters. Since the M50 is no longer available and was updated by the variants M50T and M50LT, researchers use the new versions for

^zE-mail: nicolas.stapf@ipv.uni-stuttgart.de

their work. Ai et al. (2022)¹⁶ modeled the M50T by simulating the graphite and silicon particles in the negative electrode separately using the parameters determined by Chen et al. (2020). Especially, in the transition region from the active area of graphite and silicon, the model lacks in accuracy. Besides using the parameterization from another cell, the authors state the issue of an unknown silicon mass fraction in the negative electrode. The use of new and uncharacterized cells requires new determination of the relevant parameters. This parameterization as well as the determination of the silicon mass fraction will be addressed in the present study.

Complementary to this, Heenan et al. (2020)¹¹ focused on the silicon-containing LG MJ1 and performed 3D particle imaging alongside electrochemical analysis. While this enhanced understanding of the electrode microstructure with focusing on degradation, essential parameters for electrochemical modeling (e. g. the open circuit potentials) are not provided and the silicon content was not precisely determined (range between 2.2% and 5%).

Baazouzi et al. (2023)²⁴ contributed by investigating the mechanical design aspects of the LG M50LT, including tab geometry and winding configuration through computer tomography (CT) imaging. Although informative for thermal or electrical modeling, their work does not provide detailed chemical and electrochemical analysis of the cell.

A broader cross-comparison of commercial cells in the 18650 format was carried out by Krause et al. (2021),²⁹ using X-ray diffraction (XRD), SEM, and EDS, with a focus on aerospace applications. However, their analysis focuses on performance tests to evaluate the cycle life at different temperatures and depths of discharge.

In terms of chemical degradation, Flügel et al. (2022)³⁰ utilized glow discharge optical emission spectroscopy (GDOES) to detect lithium deposition on graphite/silicon composite negative electrodes, providing important insights into lithium plating risks. Meanwhile, Schmitt et al. (2022)³¹ studied aging mechanisms in silicon-containing electrodes under consideration of the degradation-induced balancing shifting and scaling of the negative and positive electrode using the example of the LG MJ1. These works underline the challenges in the long-term stability of silicon-based composites.

Beyond characterization, Fill et al. (2024)³² focused on a model-based approach to simulate the composite open circuit potential based on the pure silicon and graphite potentials for the LG MJ1 for determining the amount of silicon present in the electrode. Heugel et al. (2024)³³ applied differential voltage analysis (DVA) to determine the silicon capacity fraction and to examine the aging behavior of a Samsung INR18650-30Q cell. These works only determine the silicon fraction based on one examination technique. By combining different techniques, which is done in this work, the silicon fraction can be determined more reliably and can be validated against each other.

A methodological advance was made by Kang et al. (2021),^{34,35} who proposed improvements to the galvanostatic intermittent titration technique (GITT) measurement to determine the diffusion coefficient of solid particles more sufficiently and to increase validity. In contrast to Chen et al. (2020),¹⁵ the present study applies these guidelines and methods from Kang et al. (2021) to determine the diffusion coefficient for the LG M50T and M50LT which should lead to more reliable measurement results.

Finally, Bonkile et al. (2024)¹⁹ incorporated the parameterization from Chen et al. (2020) for the LG M50T (although Chen et al. (2020) characterized the LG M50) and dealt with the research question whether silicon is worth it by modeling the degradation behavior of the LG M50T. They concluded that the operating conditions of silicon-containing electrodes must be controlled to not only cycle silicon due to its fast aging behavior. Thus, operating the battery cell at low SoCs (below 30%) should be avoided. In addition, avoiding lower C-rates and higher silicon contents are stated to potentially increase cycle life of silicon-containing electrodes.¹⁹ Their research question is directly linked to this study

but focuses more on the degradation behavior rather than on the battery evolution conducted by the manufacturer.

In contrast to prior work, the present study contributes by providing a detailed characterization of the two commercial cylindrical battery cells LG M50T and M50LT which are often used in research. Recent findings such as the method to determine the solid diffusion coefficient are applied and five different techniques to estimate the silicon fraction in the composite electrode of the M50T are combined to verify the results with each other. In addition to the measurement techniques applied in Chen et al. (2020), XRD and GDOES measurements are done to reveal the structural and chemical composition of the electrodes. To summarize, this work provides a direct, systematic comparison of graphite–silicon composite electrodes with and without silicon under identical conditions for characterizing the physical, chemical and electrochemical properties of the battery cells LG M50, M50T and M50LT.

Contribution and structure.—This study extends the work of Chen et al. (2020) by investigating not only the LG M50, but also its newer subvariants, the LG M50T and LG M50LT, which exhibit significant differences in electrochemical design and behavior. In the context of this work, the battery cells LG M50T and M50LT were disassembled to extract the electrode materials, enabling detailed insights into their physical, chemical, and electrochemical properties. The electrodes were appropriately prepared for further investigation in various measurement devices. The resulting data were then compared with the experimental results recorded by Chen et al. (2020),¹⁵ allowing to identify key differences and similarities between the three battery cells—LG M50, M50T, and M50LT—and characterize them in detail. These findings are essential for understanding the evolution of these cells, which were all designed by LG Chem in the 21700 format and released sequentially.

As the original LG M50 is no longer available on the market, recent research increasingly focuses on the M50T and M50LT cells (e. g.,^{16,19,24}). Our study reveals that the M50LT variant does not contain silicon in the negative electrode, unlike the LG M50, a distinction that has often been overlooked and may lead to misinterpretations in the literature. Thus, a detailed characterization of these subvariants is both timely and essential.

In addition, our study provides several new insights. SEM imaging uncovered that the positive electrode consists of agglomerated nanoparticles and carbon fibers, which likely function as conductive additives. Moreover, we applied XRD and GDOES techniques not used by Chen et al. (2020),¹⁵ offering new structural and compositional perspectives.

Our study also improves upon the determination of solid-state diffusion coefficients by adopting a more reliable approach based on the methodology proposed by Kang et al. (2021),^{34,35} which enhances the accuracy of the GITT analysis. Furthermore, while Chen et al. (2020) quantified the silicon content using EDS, a semi-quantitative method, we utilized a more robust multi-technique approach, combining EDS, GDOES, simulated composite OCP, half-cell and full-cell DVA, to more accurately quantify the silicon mass fraction in composite negative electrodes.

The resulting data can be used and post-processed to set up and parametrize electrochemical simulation models and to facilitate the interpretation of experimental results for these cells. These contributions underscore the scientific advancements and added value of our work compared to that of Chen et al. (2020).

The structure of this publication is illustrated in Fig. 1 and can be primary divided into the Sections physical, chemical, and electrochemical analysis. The respective measurement techniques and results are listed below. Both chapters “Material and Methodology” and “Results and Discussion” are structured in this way.

To improve the reading flow, the cells are called LG M50, M50T and M50LT in the following. For easier visual interpretation of the data and results in this study, the LG M50 is always marked in green

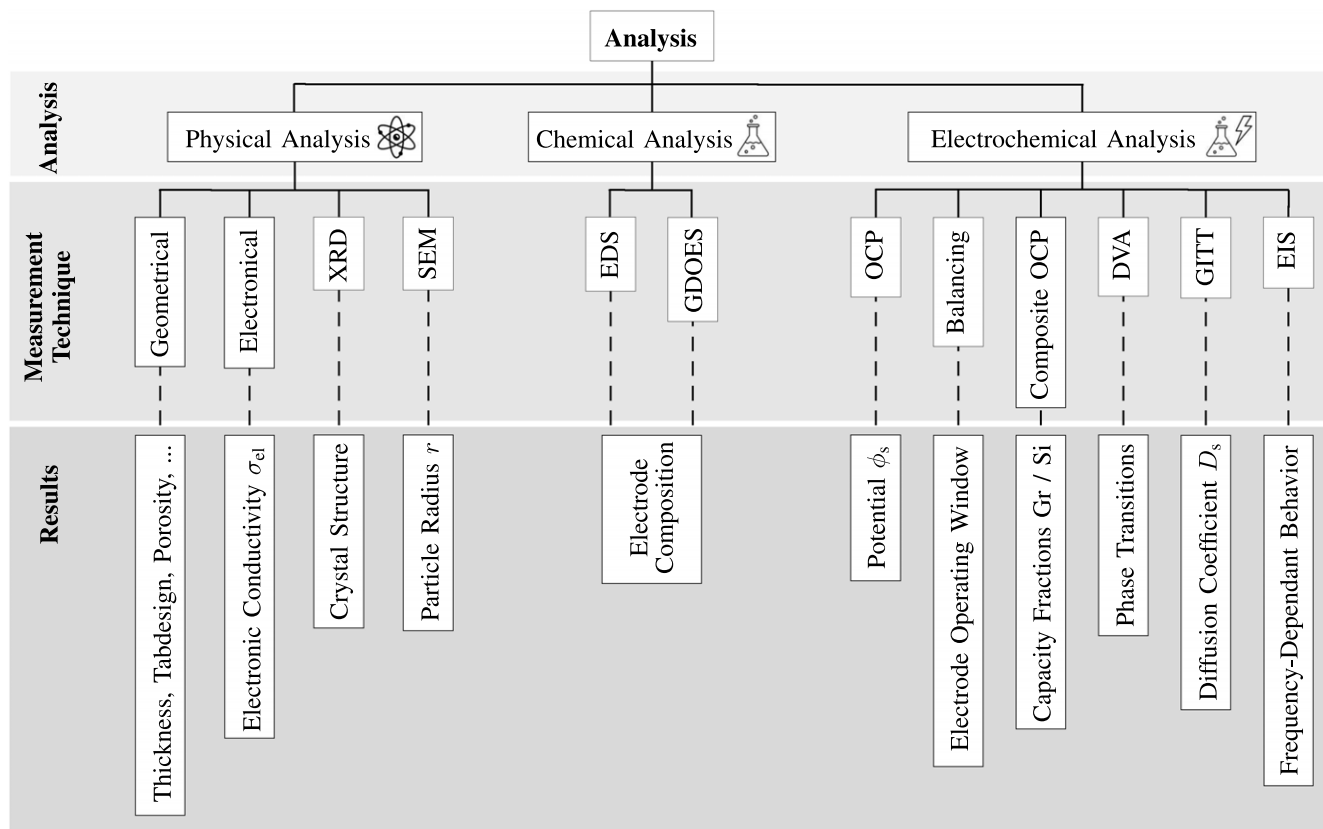


Figure 1. Structure of the analysis techniques and results covered in this study.

with rectangle markers, the LG M50T in red with circular markers and the LG M50LT in blue with diamond-shaped markers. The full-cells have a medium light color and the negative and positive electrodes have a light and dark color scheme, respectively. For the composite negative electrode of the LG M50T, graphite is displayed in light red and silicon in orange. For lithiation/charge, the curves are solid; dotted lines are used for delithiation/discharge. Simulated or calculated results are black.

Material and Methodology

The cylindrical INR21700 battery cells LG M50T and LG M50LT were investigated in this study. Each cell was analyzed in terms of their physical, chemical, and electrochemical properties and behavior. Therefore, the cells were examined in their cylindrical design and the properties of their electrode material were determined. Before analyzing the electrode materials, the cells need to be disassembled and the samples need to be prepared appropriately which is described in the first Section of this chapter. The following Sections are divided into the physical, chemical, and electrochemical analysis. The experiments were compared with the results for the LG M50 battery determined by Chen et al. (2020).¹⁵ All measurements were performed at 25°C.

As will be outlined in the results Section later, the LG M50 and M50T consist of a graphite/silicon composite negative electrode, whereas the LG M50LT only consists of a pure graphite electrode. The positive electrodes are similar for the three electrodes and can be referred to as NMC-811.

Cell disassembly and sample preparation.—Before disassembling the pristine battery cells, they were discharged with a constant current of 0.2 C to 2.5 V and then discharged with a constant voltage at 2.5 V until the current was below C/100. Subsequently, the cells were transferred to an argon-filled glove

box (water and oxygen content <0.1 ppm) and the cell cans were opened to extract the electrode materials. Directly after cutting off one terminal of the cell can, the cell is weighted, then transferred into a vacuum oven and dried for 16 hr at 50 °C. The weight difference before and after the drying process is determined as the amount of electrolyte that the cell contains. It should be noted that the conductive salt remains in the battery cell and only the solvent and additives are likely evaporated. After removing the battery can, the lateral dimensions and the thickness of the electrode sheets were firstly measured using a measuring tape and a micrometer screw, respectively, as will be described in the following section. Afterwards, the double-coated electrodes were delaminated on one side by using N-methyl-2-pyrrolidone (NMP) and a glass fiber pen to reduce artifacts in the subsequent experiments. The single-coated electrodes were punched into smaller circular coins with a diameter of 10 mm and 18 mm by using an EL-Cell EL-Cut. These sample coins were washed in dimethyl carbonate (DMC) for 2 h and dried in a vacuum oven at 50 °C for 16 h to remove conductive salt and electrolyte residues. This process is visualized in Fig. 2. Images of the cell disassembly are presented in the supplementary information in Fig. S2. After all these preparation steps, the samples were physically, chemically and electrochemically analyzed.

Physical analysis.—For determining the physical properties of the electrodes, the geometry such as the dimensions of the electrode sheets, the electronic conductivity and the X-ray diffraction were measured and scanning electron microscopy was conducted.

Geometrical measurements.—The geometrical parameters of the batteries such as the length and width of the electrode sheets, the tab position and the separator thicknesses were determined with a measuring tape and micrometer screw, respectively (see Fig. S3 in the supplementary information). The electrode

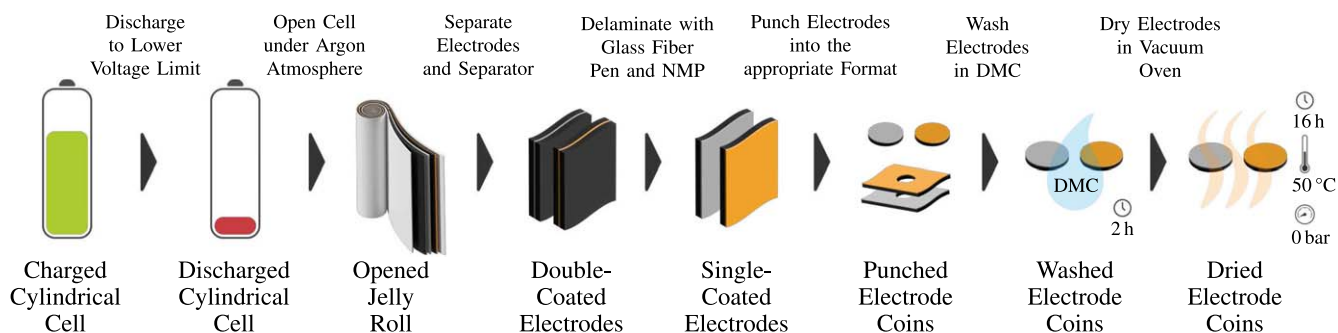


Figure 2. Cell teardown process and sample preparation.

thickness for the double-coated electrode including the current collector t_{double} was measured at ten different positions across the electrode sheet using a micrometer screw with an accuracy of $1\ \mu\text{m}$ to ensure reproducibility and account for production variations. To determine the electrode coating thickness and the current collector thickness, the electrodes were delaminated as explained in the Section before. The thickness of one electrode coating layer plus the current collector thickness t_{single} was measured again at ten different positions across the electrode. All measurements were conducted for each electrode and the results were averaged respectively. The thickness variance between the measurements for one electrode was below $\pm 2.5\ \mu\text{m}$. By using the results for the single-coated and double-coated electrodes and assuming that the coatings have the same thickness on both sides of the electrode, the electrode coating thickness t_{coating} and current collector thickness t_{CC} can be determined by

$$t_{\text{AM}} = t_{\text{double}} - t_{\text{single}} \quad [1]$$

and

$$t_{\text{CC}} = 2t_{\text{single}} - t_{\text{double}}. \quad [2]$$

The electrode loading was evaluated by using washed and dried circular double-coated coins with a diameter d of 18 mm. Five coins for each electrode were weighted with the analytical balance Kern ADB 200-4 with an accuracy of 0.1 mg. The averaged weights of the coins m_{double} and the determined thicknesses t_{AM} and t_{CC} were taken to determine the coating mass of the coin m_{coating} to

$$m_{\text{coating}} = m_{\text{double}} - \rho_{\text{CC}} \cdot \frac{\pi}{4} \cdot d^2 \cdot t_{\text{CC}} \quad [3]$$

where ρ_{CC} is the density of the current collector. For the negative and positive electrodes, densities of $\rho_{\text{CC,neg}} = \rho_{\text{Cu}} = 8.96\ \text{g cm}^{-3}$ and $\rho_{\text{CC,pos}} = \rho_{\text{Al}} = 2.70\ \text{g cm}^{-3}$ were used, respectively.³⁶ The electrode loading ρ_{A} and the density of the coating ρ_{coating} were calculated by

$$\rho_{\text{A}} = \frac{m_{\text{coating}}}{2 \cdot \frac{\pi}{4} \cdot d^2} \quad [4]$$

and

$$\rho_{\text{coating}} = \frac{\rho_{\text{A}}}{t_{\text{coating}}} \quad [5]$$

Besides the active materials, the electrode coatings also consist of additive such as carbon black and binder. Since the mass fractions of additives and binders are usually kept low in commercially available battery cells to increase the specific energy density, their contributions (e.g. the change in the crystal density of the coating) are neglected in the following. Consequently, the electrolyte volume fraction ϵ_{el} can be calculated by

$$\epsilon_{\text{el}} = 1 - \frac{\rho_{\text{coating}}}{\rho_{\text{crystal}}} \quad [6]$$

where ρ_{crystal} is the crystal density of the active material, i.e. the theoretical density of the active material components for a porosity of 0%. The crystal density ρ_{crystal} for NMC-811 is $4.95\ \text{g cm}^{-3}$.¹⁵ Since the density of graphite ($2.26\ \text{g cm}^{-3}$,^{15,37}) and silicon oxide ($2.25\ \text{g cm}^{-3}$ for $\text{SiO}_{1.6}$,³⁸) are similar and the mass fraction of graphite is much higher than that of silicon oxide (as will be described in the chemical and electrochemical analysis), a crystal density for both the composite and pure graphite negative electrodes of $2.26\ \text{g cm}^{-3}$ was used.

The tortuosity τ_{el} can be calculated using the following expression

$$\tau_{\text{el}} = \epsilon_{\text{el}}^{-b} \quad [7]$$

where b is the Bruggeman coefficient. Assuming spherical particles, the Bruggeman coefficient b is 0.5.³⁹

Electronic conductivity.—To measure the electronic conductivity of the electrode σ_{coating} a four-point probe measurement method was used. The setup of the test stand is described in Schneider et al.⁴⁰ A table multimeter Rigol DM3058E was used in the resistance mode which applies a current of 1 mA and measures the resistance with an accuracy higher than 0.035%.

For measuring the electronic conductivity of the active material only, the active material was removed from the current collector with an adhesive tape. In addition, the tape also acts as an insulating layer. The thickness of the removed active material $t_{\sigma,\text{el}}$ was measured at ten different location and was then averaged. The measurements were performed inside a glove box under argon atmosphere to prevent contamination, decomposition or oxidation of the active materials and the SEI. The measured resistance can be converted into the electronic conductivity σ_{coating} by

$$\sigma_{\text{coating}} = \frac{\ln(2)}{\pi \cdot t_{\sigma,\text{el}} \cdot R_{\text{el}}} \quad [8]$$

where R_{el} is the measured resistance.⁴¹ The electronic conductivity σ_{coating} is an effective value since it does not account for the porosity of the electrode.

XRD measurements.—A PANalytical EMPYREAN with Cu radiation was used in the Bragg-Brentano mode for the XRD analysis to investigate the crystal structure of the electrodes. For the XRD measurements the measured 2θ angle range was 10° to 70° . The angle change per step was set to 0.006565° and the measuring time per decade to 90 s. The used primary and secondary mask had a size of 2 mm and 6 mm, respectively. During the measurement the sample was spinned with a revolution time of 1 s. The pulse height distribution range for the negative and positive

electrode was between 25% and 80% and between 48% and 80%, respectively. Data was analyzed using the database PDF5+ of the international centre for diffraction data (ICDD).

SEM measurements.—An SEM from JEOL (JSM-IT800) was used to illustrate the electrode particles and determine their radius. For this, a secondary electron detector was used for the top-view images to obtain information about the surface (e. g. about the topology). An accelerating voltage of 1 kV, 3 kV and 5 kV was applied thereby. After the sample preparation in an argon-filled glove box, the samples were transferred in a hermetic transfer vessel to the SEM without being contaminated with ambient air.

Chemical analysis.—The extracted and prepared electrode coins from the commercial electrodes were examined in the chemical analysis. EDS and GDOES were performed to evaluate the elemental composition of the samples. The results from the different measurement techniques were later quantitatively compared with one another to avoid misinterpretation.

EDS measurements.—In combination with the above mentioned SEM from JEOL, EDS was conducted for each electrode of the LG M50T and M50LT to determine the elemental composition of the electrode and to connect the particle appearance with the elemental composition on the surface of the samples. EDS was performed at an accelerating voltage of 15 kV. The same samples were used for the SEM and EDS measurements.

GDOES measurements.—To further investigate the elemental composition of the samples on a larger area compared to SEM/EDS and along the electrode thickness, the GDOES device GDA750HR by Spectrumba was used with a sputtering diameter of 2.5 mm. The setting parameters for the negative electrodes were defined similarly to.³⁰ A discharge voltage of 550 V, a carrier pressure of 2 hPa and a frequency of 2501 Hz in the radio frequency (RF) mode with a duty cycle of 15% were specified. For the positive electrodes, the discharge voltage was set to 770 V, the carrier pressure to 3 hPa and the frequency to 513 Hz in the RF mode with a duty cycle of 35%. The used sputtering gas was argon. The element specific emission lines used for the analysis were: H (121.6 nm), O (130.2 nm), C (165.7 nm), P (178.3 nm), S (180.7 nm), Si (288.2 nm), Cu (327.4 nm), Ni (341.4 nm), Co (345.4 nm), Al (396.2 nm), Mn (403.4 nm), Li (670.7 nm). The samples were transferred from the argon-filled glove box to the GDOES device in a transfer vessel to guarantee that the samples were not contaminated with ambient air. The sample area and transfer vessel are presented in the supplementary information in Fig. S6.

Electrochemical analysis.—For determining the electrochemical behavior of the cells, the extracted electrode materials from the commercial cells were used to set up experimental cells in PAT-Cells from EL-Cell. With this setup, half-cells (negative or positive electrode from the commercial cells vs lithium metal) were built up as three-electrode-setups with a lithium-coated reference ring. The experimental setup can be seen in the supplementary data in Fig. S7. The analyzed potential was always taken from the reference ring to exclude the overpotentials occurring on the lithium metal electrode. The single-coated electrodes had a circular diameter of 18 mm and the glass fiber separator Whatman GF/A with a thickness of 260 μm was used. A mixture of 1M LiPF₆ in fluoroethylene carbonate (FEC) and diethyl carbonate (DEC) with a volumetric ratio of 2:8 was used as the electrolyte.

After setting up the experimental cells, the cells were first cycled with a constant current of C/3 for three cycles as a formation procedure between the newly added electrolyte and the commercial electrode. The C-rate was determined based on the area ratio of the cylindrical and experimental cell. For cycling the experimental cells, a BaSyTec CTS Lab battery tester was used with a voltage accuracy

of 1 mV and a current accuracy of 0.2 μA . An ACT0550 battery tester from PEC Corp. with a voltage accuracy higher than 1 mV and a current accuracy of 60 μA was used for cycling the cylindrical cells.

As part of the electrochemical analysis, the pseudo open circuit potentials (pOCPs), pseudo open circuit voltages (pOCVs), and the electrode balancing were determined. In addition, simulations of the composite OCP, differential voltage analysis (DVA), GITT measurements, and electrochemical impedance spectroscopy (EIS) were conducted. The following Sections detail the methodologies employed for these techniques.

Open circuit potential and voltage.—By using the experimental cells, the pOCPs of the negative and positive electrodes were measured at a constant current of C/100 for lithiation and delithiation. The negative electrodes were cycled in a voltage range of 0.005 V to 1.3 V and the positive electrodes in a range of 3.3 V to 4.3 V. The ranges are chosen based on the operating window of the full-cell examined in Section Electrode Balancing and based on the nearly fully lithiated states of the electrodes.⁴²

In addition, the pOCVs of the cylindrical cells were determined with a cycling current of C/100 for charge and discharge in a voltage window between 2.5 V and 4.2 V.

Electrode balancing.—Taking the half-cell pOCPs of the negative and positive electrodes of the appropriate battery cell, the measured pOCV of the pristine cylindrical cells can be reproduced by combining the half-cell pOCPs properly. For this, the measured and normalized pOCPs need to be linearly scaled and shifted to fit the SoC range of the cylindrical cell. The transformation from half-cell SoC $x_{\text{half},i}$ to full-cell SoC x_{full} can be done by

$$x_{\text{full}} = \alpha_i \cdot x_{\text{half},i} + \beta_i \quad [9]$$

where i is either the negative or positive electrode, α is the linear scaling factor and β is the linear shifting factor of the half-cell pOCP curve.³¹ For the negative electrode, α_{neg} can be understood as the oversize of the negative electrode capacity compared to the positive electrode capacity for e. g. reducing plating at high full-cell SoCs. To determine the appropriate values of α_i and β_i the following equation is optimized by finding the minimum root mean square error (RMSE) between measured and calculated voltage. Thereby, the electrode potentials are described as a function of the full-cell SoC x_{full} ³¹

$$V_{\text{full}}(x_{\text{full}}) = \phi_{\text{pos}} \left(\frac{x_{\text{full}} - \beta_{\text{pos}}}{\alpha_{\text{pos}}} \right) - \phi_{\text{neg}} \left(\frac{x_{\text{full}} - \beta_{\text{neg}}}{\alpha_{\text{neg}}} \right) \quad [10]$$

V_{full} , ϕ_{neg} and ϕ_{pos} are the pOCV of the cylindrical cells and the pOCPs of the appropriate negative and positive electrodes, respectively.

Simulation of the composite OCP.—The composite OCP of the graphite/silicon oxide composite electrode of the LG M50T is simulatively calculated to determine the capacity fraction of silicon in the composite electrode. For graphite, the potential data from⁴³ was extrapolated for potentials above 0.4 V (smaller than a relative capacity of 0.05) with data from.⁴⁴ Since only pure silicon participates reversibly in the alloy reaction with silicon,¹⁶ the OCP of pure silicon from⁴⁵ was used. Due to the significant hysteresis of silicon between lithiation and delithiation, the appropriate OCPs for the lithiation and delithiation processes were simulated separately for graphite, silicon and the composite OCP. Since thermodynamic equilibrium should be given at every potential in an OCP, the graphite and silicon particles have the same potential inside the

composite electrode. Based on that, the potential curves of pure graphite and silicon can be combined and the composite potential curve can be created. Starting at fully lithiated or delithiated state, a discrete capacity fraction is charged into or discharged from the electrode until the fully delithiated or lithiated state is reached. For every (de-)lithiation step it needs to be decided whether graphite or silicon particles are (de-)lithiated based on the current potential and lithiation state. The simulated results were validated with the measured pOCP of the LG M50T.

Differential voltage analysis.—To identify characteristics in the pOCPs, the derivatives of the pOCPs were investigated in differential voltage analysis (DVA) and incremental capacity analysis (ICA). Therefrom, the capacity fraction of silicon in the composite electrodes can be estimated and the phase transitions of the active materials observed. The DVA was also conducted for the cylindrical cell pOCV and the results were compared with the half-cell pOCPs.

To determine the derivatives, the capacity or voltage were discretized into 4000 data points and the results were filtered using the moving average over 50 data points.

Galvanostatic intermittent titration technique.—In 2021, Kang et al. analyzed the applied methods prominent in literature for the galvanostatic intermittent titration technique for determining the diffusion coefficient of the solid electrode material D_s .^{34,35} The authors identified possible error sources and developed a modified method of how to implement GITT properly and consequently measure the diffusion coefficient D_s more reliably.

In this study, GITT was applied according to Kang et al. (2021) to determine the diffusion coefficients of the solid electrode material D_s . Hence, short and constant current pulses were induced on experimental half-cell level and the relaxation behavior under an open-circuit condition was analyzed during the GITT experiment. The short current pulses result in a change in the electrochemical composition on the surface of the active material particles inside the electrode which leads to a relaxation behavior related to the diffusion coefficient D_s . Kang et al. (2021) suggest using short pulses relative to the sample capacity so that the shift in SoC of the electrode can be neglected. By analyzing the relaxation instead of the pulse itself, overpotential problems can be avoided.^{34,35}

Furthermore, the relaxation time variable $\sqrt{t_{\text{relax}} + \tau} - \sqrt{t_{\text{relax}}}$ was used where t_{relax} is the relaxation time and τ is the galvanostatic pulse duration. The relaxation time t_{relax} is 0 when the current pulse ends and the relaxation starts ($t - \tau = t_{\text{relax}}$).

When plotting the relaxation voltage of each current pulse over the time domain $\sqrt{t_{\text{relax}} + \tau} - \sqrt{t_{\text{relax}}}$, a linear region can be identified. Since diffusion is present at longer time scales, this linear region can be referred to the diffusion process occurring in the electrode particles.³⁵ As an example, the relaxation voltage over the relaxation time variable is illustrated in Fig. 3 for one current pulse including the identified linear region.

The linear region is reciprocal proportional to the diffusion coefficient D_s and used for further processing. The linear gradient of the relaxation was evaluated in a voltage over $\sqrt{t_{\text{relax}} + \tau} - \sqrt{t_{\text{relax}}}$ plot for each current pulse. To interpret the plot in the real time scale, the data needs to be read starting from the right to the left. The non-linear slope between $5 \text{ s}^{0.5}$ and $6 \text{ s}^{0.5}$ can be referred to surface effects of the particles occurring directly after the pulse. In order to identify the linear relaxation region correctly, the current pulse length τ should fulfill the following condition for spherical systems

$$\hat{\tau} = \frac{D_s \cdot \tau}{r^2} \leq 0.02. \quad [11]$$

where $\hat{\tau}$ is the dimensionless time scale and r is the radius of the electrode particles.³⁵

To calculate the diffusion coefficient in the solid electrode material D_s the following equation can be used assuming that the

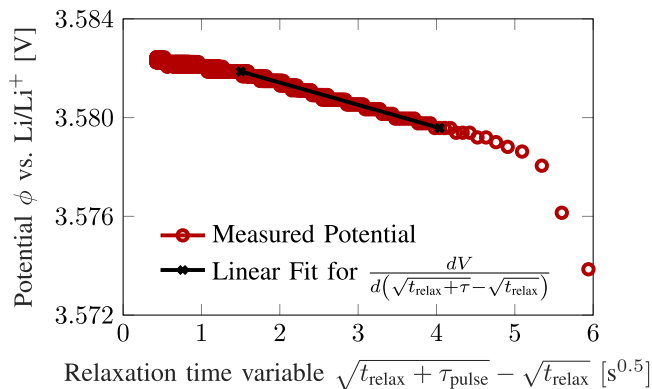


Figure 3. GITT lithiation pulse 5 for the positive electrode of the LG M50T in the time domain $\sqrt{t_{\text{relax}} + \tau} - \sqrt{t_{\text{relax}}}$ with extracted linear region used for determining the diffusion coefficient D_s .

electrode consists of spherical particles⁴⁶

$$D_s = \frac{4}{9\pi} \left(\frac{r \cdot \Delta V_{\text{eq}}}{\tau \cdot \frac{dV}{d(\sqrt{t_{\text{relax}} + \tau} - \sqrt{t_{\text{relax}}})}} \right)^2. \quad [12]$$

ΔV_{eq} is the voltage difference before the current pulse and after the relaxation period. $dV / d(\sqrt{t_{\text{relax}} + \tau} - \sqrt{t_{\text{relax}}})^{-1}$ can be extracted for each current pulse from the linear slope in the $\sqrt{t_{\text{relax}} + \tau} - \sqrt{t_{\text{relax}}}$ domain as shown in Fig. 3. For the composite electrode of the LG M50T, the particle radius r is calculated by the average of the graphite and silicon oxide particles weighted by their volume fraction. The volume fraction of graphite and silicon oxide is nearly equivalent to their mass fraction since their densities are similar (see Section before).

The GITT measurements were performed with the experimental half-cell three-electrode setup with a lithium-coated reference ring outlined above. The measured lithium reference potential was used for the analysis. An adequate magnitude of the current pulse was found to be C/15. That means it was large enough to show a measurable change in the voltage and small enough to not introduce unintended error sources into the GITT experiment. Since the parasitic current occurring in the GITT measurement is usually very low,³⁵ it was neglected in this study.

During one GITT experiment 25 current pulses were induced with a pulse duration τ of 36 s. After each pulse and subsequent relaxation period, the SoC was ramped up with C/10 by about 4% per step. The GITT experiment was initialized by reaching the lower voltage limits of the electrodes with C/100. For the negative electrode a delithiation pulse and for the positive electrode a lithiation and delithiation pulse were applied. The relaxation periods before and after the current pulses were set to 8 h.

During the ambipolar chemical diffusion, lithium ions and electrons are transported in the bulk. Using the GITT measurements, both the chemical diffusivity and the ionic conductivity can be determined.³⁵ Assuming that the ions—and not the electrons—limit the chemical diffusion, the ionic conductivity σ_i can be determined with

$$\sigma_i = \frac{D_s \cdot F^2}{v_m \cdot \frac{\Delta V_{\text{eq}}}{\Delta x}} \quad [13]$$

where $\Delta V_{\text{eq}} \Delta x^{-1}$ is the difference in the open circuit voltage V_{eq} divided by the difference in stoichiometrical composition x ($x \in [0, 1]$) during one pulse.^{35,47} The molar volume v_m of one chemical species (Li_xC_6 , $\text{Li}_{15x}\text{Si}_4$ and $\text{Li}_{1-x}\text{Ni}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$) and the Faraday constant F are taken. For the composite electrode, the molar volume v_m is determined as the average molar volume weighted by the volume fraction of graphite and silicon oxide.

The validity of the conditions and assumptions made for the GITT experiments (i.e. the pulse duration and that the ions limit the chemical diffusion) will be proofed in the Section results and discussion.

Electrochemical impedance spectroscopy.—To ascertain the parameters for an equivalent circuit model, potentiostatic EIS was carried out on electrode level. Therefore, the experimental cells were analyzed with the combination of the BaSyTec battery tester and a Gamry Reference 3000AE suitable for EIS measurements. After each GITT current pulse and subsequent relaxation period, an EIS measurement was conducted. The cells were relaxed for 8 h after performing the EIS measurement. A frequency range from 100 kHz to 10 mHz was applied and the root mean square value of U_{ac} was set to 10 mV while 10 points per decade were recorded.

Results and Discussion

The results of the measurement techniques outlined in the Section before are presented and discussed in the following. The M50T and M50LT were analyzed in this study, and the results for the M50 were taken from Chen et al. (2020).¹⁵ This chapter is structured analogously to the previous chapter: The physical, chemical, and electrochemical analysis are treated successively.

Physical analysis.—First, the geometrical measurements are depicted. Second, the electronic conductivity σ_{coating} is calculated and third, XRD measurements are shown.

Geometrical measurements.—The geometrical measurements and calculations are summarized in Table I including also the data from Ref. 15 for the LG M50. It is noticeable that the length of the electrode of the LG M50 and M50T are similar whereas the M50LT is significantly larger. The total coating length of each electrode of the M50LT is approximately 25 cm longer than that of the M50 or M50T. The coating length of the double-coated electrode is the same

on both sides for all positive electrodes. The negative electrodes are about 6 cm shorter on one coating side due to the geometry of the jelly roll with the negative electrodes forming the outer layer. However, the negative and positive electrode coating thicknesses t_{coating} of the M50LT are about 14% and 21% smaller than that of the other cells. The coating thicknesses are nearly identical for the M50 and M50T (less than 2 μm difference). The thinner electrode of the M50LT leads to smaller diffusion paths for the ions and electrons and therefore higher currents should be applicable. Furthermore, the concentration gradient of lithium through the thickness of the electrode is likely reduced, particularly at high currents, potentially due to the thinner electrode design; however, other factors such as porosity, tortuosity, and electrolyte properties may also contribute and warrant further investigation. Nonetheless, according to the data sheets, the discharge currents are allowed to be twice as high for the M50T and M50LT compared to the LG M50 but the discharge currents for the M50T and M50LT are specified identically. The charge currents are defined the same for all cells.

The width of the electrodes is the same for all of the cells. By considering the total coating volume as the product of the total coating area and coating thickness, it is striking that the negative electrode of the M50LT is by 26.8% larger than the positive electrode. For the M50T and M50, the volume difference is only 18.1% and 14.8%, respectively. This influences the electrode balancing of the negative and positive electrode and will be outlined in the appropriate section.

The results of the electrode loading ρ_A according to (4) is almost identical for the LG M50 and M50T but differs from the LG M50LT. These circumstances can be traced back to the different electrode thicknesses of the batteries since the electrode coating densities ρ_{coating} calculated by (5) are almost the same for the three cells.

The electrolyte volume fraction ϵ_{el} of the positive electrode—also known as the porosity—determined with (6) is nearly identical. Only the electrolyte volume fraction of the negative electrode of the M50 is about 5%–6% lower than that of the M50T and M50LT

Table I. Summary of the determined physical and electrochemical analysis, i.e. the geometrical parameters, electronic conductivity, mean particle radii and diffusion coefficients for the LG M50, LG M50T and LG M50LT.

Parameter	Unit	LG M50 ^{a)}		LG M50T		LG M50LT	
		Neg. Electrode	Pos. Electrode	Neg. Electrode	Pos. Electrode	Neg. Electrode	Pos. Electrode
Length (outer/inner)	cm	77.5/83.5	79.0/79.0	81.5/87.5	80.8/80.8	94.1/99.0	93.7/93.7
Width	cm	6.5	6.5	6.5	6.5	6.5	6.5
Current Collector Thickness t_{CC}	μm	11.7	16.3	13.5	15.4	12.5	15.0
Electrode Coating Thickness t_{coating}	μm	85.2	75.6	86.7	76.8	74.4	60.4
Total Coating Length	cm	161.0	158.0	169.0	161.6	193.1	187.4
Total Coating Area	cm^2	1 046.5	1 027.0	1 098.5	1 050.4	1 255.2	1 218.1
Total Coating Volume	mm^3	8 916.2	7 764.1	9 524.0	8 067.1	9 332.0	7 357.3
Electrode Loading ρ_A	mg cm^{-2}	14.85	24.69	14.01	26.03	11.96	20.47
Electrode Coating Density ρ_{coating}	g cm^{-3}	1.743	3.266	1.615	3.390	1.608	3.389
Electrolyte Volume Fraction ϵ_{el} based on Theoretical Crystal Density	%	23.00	32.00	28.54	31.52	28.83	31.54
Tortuosity τ_{el}	—	2.085	1.768	1.872	1.781	1.862	1.781
Tab Position	cm	—	—	0	62.6	0	63.0
Separator Thickness	μm	12.0		16.7		14.2	
Electrolyte Quantity	g	—		3.278		3.171	
Area-Specific Capacity	mAh cm^{-2}	5.07	4.70	5.21	5.15	4.30	4.35
Volume-Specific Capacity	mAh cm^{-3}	595.1	609.4	601.4	670.9	578.3	719.8
Mass-Specific Capacity	mAh g^{-1}	341.4	186.6	372.4	197.9	359.7	212.4
Electronic Conductivity of the Coating σ_{coating}	S m^{-1}	215.00	0.18	253.71	5.88	207.99	9.21
Mean Particle Radius r (Gr/SiO _x for composite electrodes)	μm	5.86/1.52	5.22	6.18/2.31	5.14	5.69	4.50
Mean Diffusion Coefficient $D_{\text{s,mean}}$	$\text{fm}^2 \text{s}^{-1}$	1.740	1.480	4.927	2.133	0.332	0.401

a) Data taken from Ref. 15.

calculated with the same method. The higher porosity for the latter mentioned cells results in a lower active material content in the electrodes. However, the porosity determined with a focused ion beam scanning electron microscope (FIB-SEM) by Ref. 15 was 27% which is in the range of the values calculated in this study. The discrepancies can be attributed to measurement inaccuracies.

The tortuosity τ_{el} was calculated with (7) assuming spherical particles ($b = 0.5$) which is proportional to the porosity. In addition, the current collector thicknesses are nearly identical for the three batteries, whereby the current collectors of the negative electrodes are always thinner compared to those on the positive electrode sides. Due to the higher electrical conductivity of copper used as the negative current collector ($\sigma_{Cu} = 58 \text{ MS m}^{-1}$ ⁴⁸) compared to aluminum used as the positive current collector ($\sigma_{Al} = 37 \text{ MS m}^{-1}$ ⁴⁹), the negative current collector can be designed thinner maintaining the same resistance. Moreover, the density of copper is higher than that of aluminum. A thicker current collector on the negative electrode side would therefore lead to a lower energy density achievable for the battery cell. The separator thickness is within $\pm 2.5 \mu\text{m}$ similar for the three cells.

The tab position was measured from the outer edge of the jelly roll for the M50T and M50LT. The position for both cells is identical. However, it must be noted that the electrode length of the M50LT is larger so that the remaining part of the electrode which is positioned in the inner part of the jelly roll is larger as well. The width of the uncoated area of the M50T and M50LT at the position of the tab was 10 mm. The tabs itself have a thickness of 3.5 mm for both the negative and positive electrodes of the M50T and M50LT. All electrodes used a single tab design.

The weight difference before and after drying the opened cell can in a vacuum oven results in an approximate electrolyte quantity of 3.278 g and 3.171 g for the M50T and M50LT, respectively. The conductive salt is neglected in this investigation since it likely remains in the cell can. According to Qiao et al. (2022), the mass of the salt in the electrolyte is about 12%.⁵⁰ It is assumed that the residual electrolyte components have evaporated during drying. The differences in electrolyte quantity between the M50T and M50LT are likely due to manufacturing and measuring tolerances.

Electronic conductivity.—The electronic conductivity σ_{coating} was determined by (8) and summarized in Table I. For the M50T and M50LT, the order of magnitude for σ_{coating} is the same. The electronic conductivity of the negative electrodes are significantly higher (factor of 23 to 43) than those of the positive electrodes. Comparing the measured results with those from Chen et al.¹⁵ for the M50, the electronic conductivity of the negative electrode is in the same range as the M50T and M50LT. The conductivity of the positive electrode is significantly lower for the M50 compared to the M50T and M50LT. Reasons could be the different currents applied in this study (1 mA) and by Chen et al. (0.03 mA).

XRD measurements.—The diffractograms for the negative and positive electrodes of the M50T and M50LT are shown in Fig. 4. Comparing the diffractograms of the negative electrodes of the M50T and M50LT (Fig. 4a), characteristic peaks for graphite and copper are present for both electrodes. The peak at $2\theta \approx 24^\circ$ can be referred to the binder (e.g. PVDF).²⁹ Differences in the spectra can be observed at $2\theta \approx 19^\circ$ and $2\theta \approx 21^\circ$ which is illustrated in detail in Fig. 4b. Those peaks are only shown for the M50T. As analyzed with the database PDF5+ of the ICDD, the first peak was identified to be LiSi and the second peak derives from silicon dioxide SiO_2 . The characteristic peaks for SiO_2 and its reaction products with lithium only exist for the LG M50T negative electrode which implies that the M50LT does not contain silicon as an active material.

The diffractograms for the positive electrodes (Fig. 4c) are similar for both electrodes concluding that the positive electrodes might consist of a similar material composition. According to the database, the diffractograms pertain to NMC-811 and the labelling

refers to NMC-811. However, the peak at $2\theta \approx 26^\circ$ was also seen in Ref. 51 identified as graphite used as a conductive additive.

SEM measurements.—First, top view SEM images of the negative and positive electrodes of the LG M50T and M50LT were taken at different magnitudes. The images presented in Fig. 5 show a field of view of $128 \times 96 \mu\text{m}$ and were taken with a secondary electron detector, an acceleration voltage of 1 kV and at a working distance of 10 mm. Similarly to Chen et al.¹⁵ for the M50, the negative electrode particles of the M50T (Fig. 5a) and M50LT (Fig. 5b) are characterized by their angular appearance, their layered structure and large range of particle sizes. Conversely, the positive electrode particles are nearly spherical in shape and have a smaller range in particle size.

Second, close-up images (Fig. 6) were taken to show the appearance of one electrode particle by keeping the settings identical to the wide shots. For the positive electrodes and the M50T negative electrode a field of view of $21 \times 16 \mu\text{m}$ and for the M50LT negative electrode a field of view of $32 \times 24 \mu\text{m}$ was chosen. The negative electrodes (Figs. 6a and 6b) show a layered and edgy structure also shown in Ref. 15 for the M50. The focus of the negative electrode of the M50T was set to smaller particles which will later be identified as a silicon oxide (SiO_x) particle by using EDS. These particles are more spherical in shape compared to graphite but they are still angular.

The positive electrode particles (Figs. 6c and 6d) are almost spherical and one micro-particle (a few micrometers in size) consists of thousands of agglomerated smaller nano-particles (a few hundred nanometers in size). The agglomerated micro-particles are surrounded by fibers in the nano-scale range and are identified as carbon by EDS which will be outlined later. Comparing this result with the XRD diffractograms, the fibers can be referred to graphite as a conductive additive in the positive electrodes. To further analyze the carbon fibers and the nano-particles present in the positive electrodes, the ultra-high definition detector of the SEM was used. Since there is no difference between the NMC particles of the M50T and M50LT visible, Fig. 7 only shows the images for the LG M50T. In Fig. 7a, the nano-particles are roundish and surrounded by fibers. The fibers are randomly arranged and connect the nano-particles with one another. They are about 20 nm thin and 500 nm long.

Using the SEM images from Fig. 5 and the software ImageJ, the particle size distributions of the electrodes were determined. For each particle, the minimum and maximum radii were measured and the mean particle radii r of the electrodes were calculated. The particle size analysis of the composite negative electrode of the M50T was subdivided in graphite and silicon oxide particles using EDS. The particle size distribution of the active materials can be seen in Fig. 8. For the negative electrodes (Figs. 8a and 8b), the particle radii range from approximately $2.5 \mu\text{m}$ to $12.5 \mu\text{m}$ which is larger compared to the ranges of the positive electrodes ($1.5 \mu\text{m}$ to $9.0 \mu\text{m}$). The histogram in Fig. 8a shows that the particle radii of the silicon oxide particles are much smaller compared to the graphite particles. The particle size distributions resembles a Gaussian distribution for all electrodes. The range in particle size for the positive electrode of the M50T (Fig. 8c) is slightly larger compared to the positive electrode of the M50LT (Fig. 8d).

The mean particle radii r are noted down in Table I. A detailed analysis of the minimum and maximum particle sizes are outlined in the supplementary information in Table SI. The average particle radii for graphite in the negative electrodes is the highest for the active material particles. The radius of the negative electrode of the M50T is by about 5% bigger compared to that of the M50. The M50LT shows the smallest particle radius for the negative electrode which is about 3% smaller than that of the M50. The silicon oxide particles are the smallest of all active material particles with a mean particle radius of $1.52 \mu\text{m}$ and $2.31 \mu\text{m}$ for the M50 and M50T, respectively. It should be noted at this point that the lithiation state of

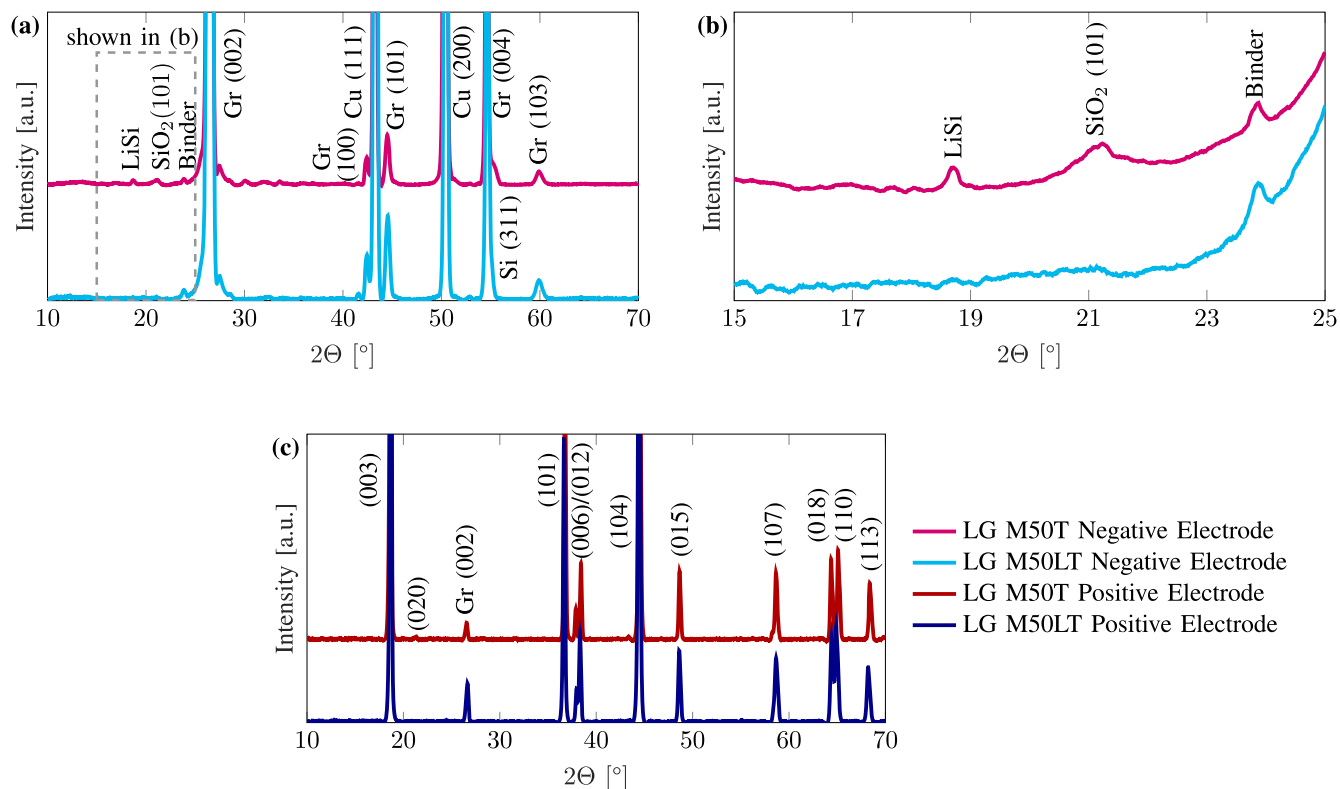


Figure 4. X-ray diffractograms for the LG M50T and M50LT for the negative (a) and positive electrodes (c) for a 2θ range of 10° to 70°. Marked rectangle in (a) indicates the zoomed 2θ region in (b) from 15° to 25°. Labelings of the peaks were taken from Refs. 29, 51–56 and 57.

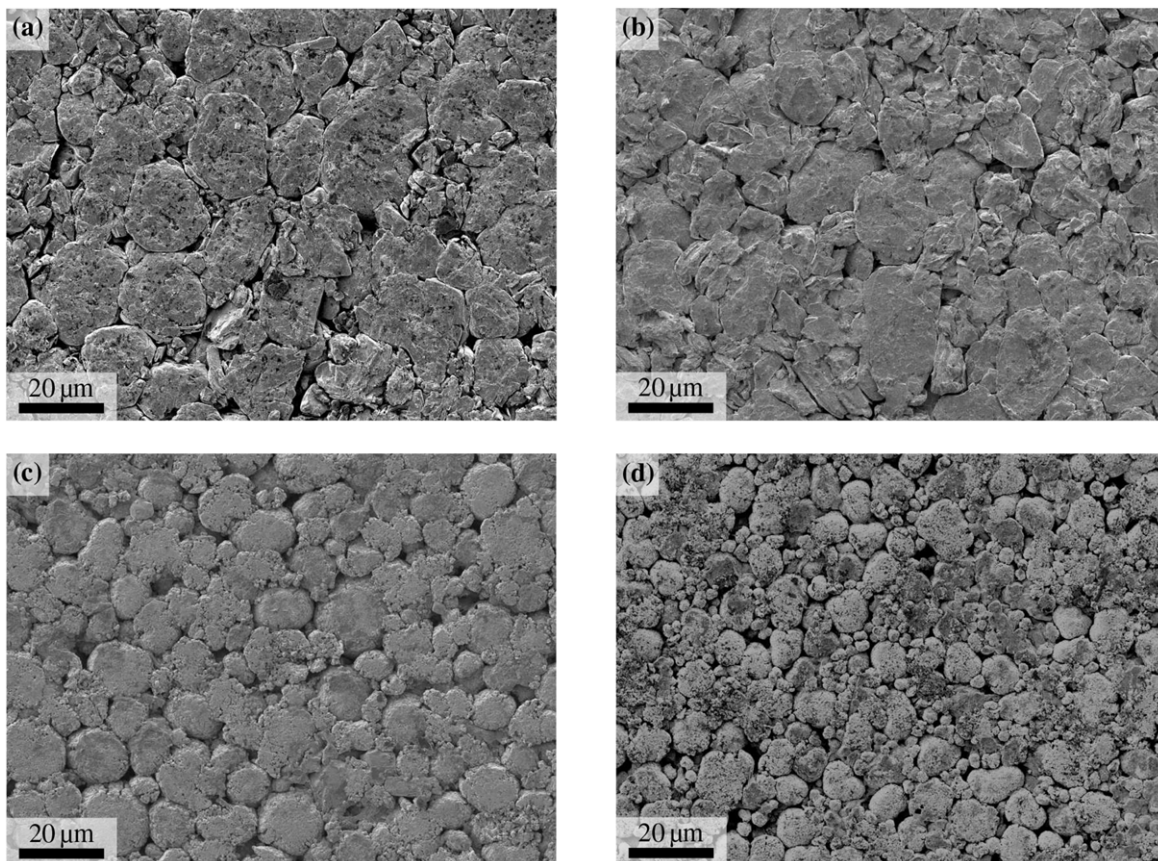


Figure 5. SEM images of the negative electrodes of the LG M50T (a) and M50LT (b) and of the positive electrodes of the M50T (c) and M50LT (d).

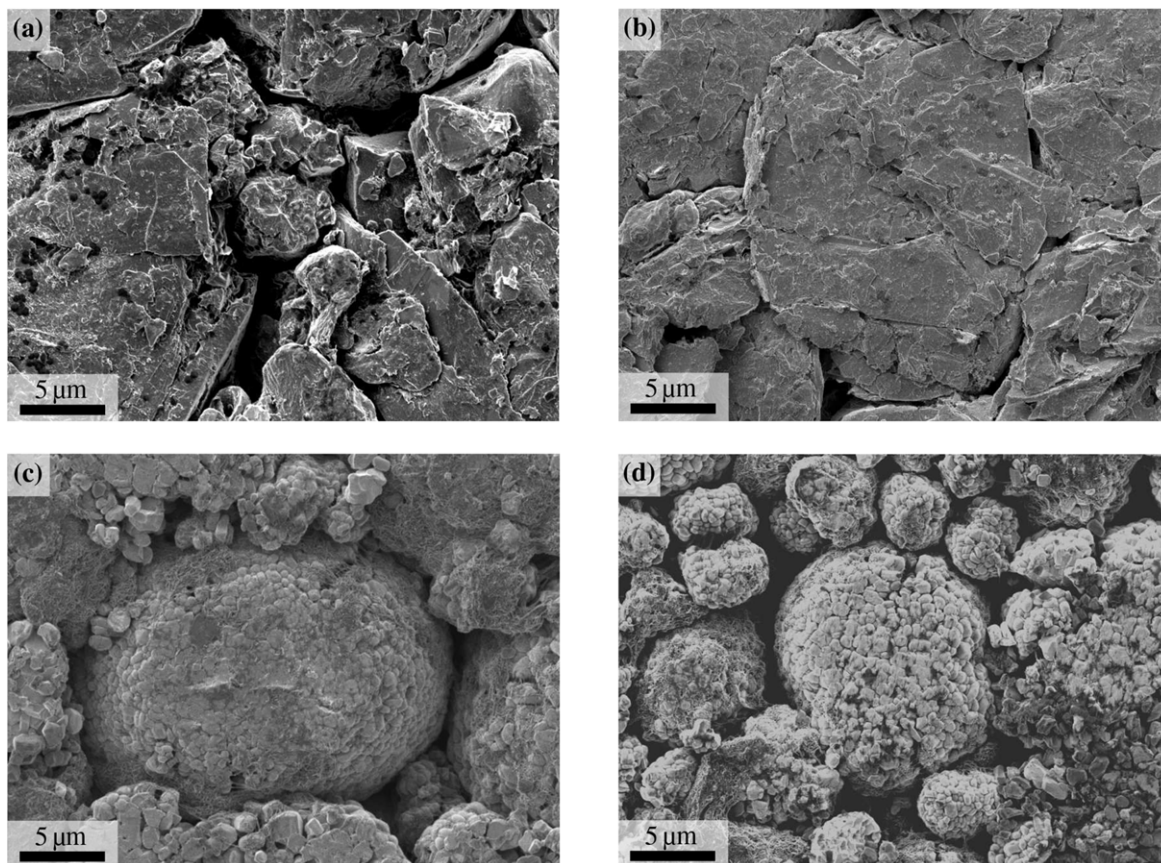


Figure 6. High magnitude SEM images of the negative electrodes of the LG M50T (a) and M50LT (b) and of the positive electrodes of the M50T (c) and M50LT (d).

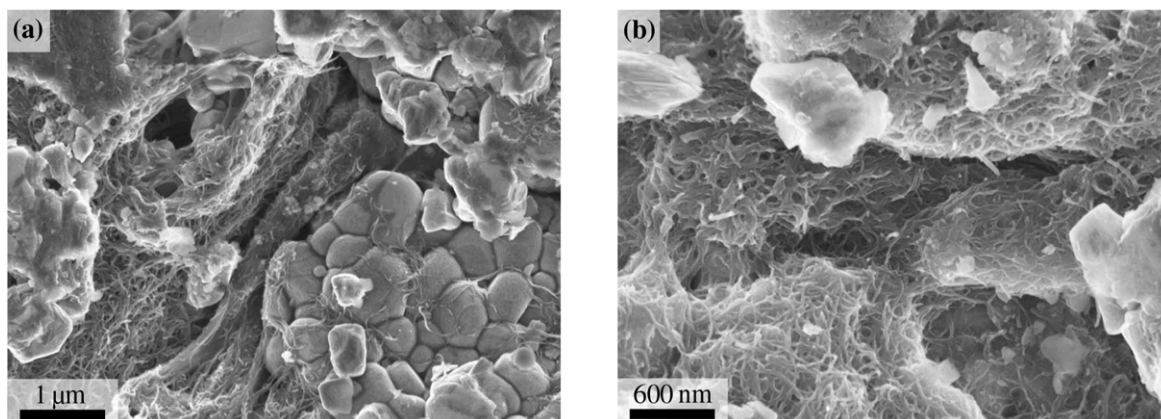


Figure 7. SEM images of the LG M50T positive electrode using the ultra-high definition detector. Field of view of $6.4 \times 4.8 \mu\text{m}$ (a) and field of view of $3.9 \times 2.9 \mu\text{m}$ (b).

the negative electrode particles is almost 0% since the electrodes were investigated after disassembling the cylindrical battery at a full-cell voltage of 2.5 V referring to a negative electrode potential above 1 V (as will be outlined in Section Electrode Balancing). The volume of the particles is therefore the lowest. Especially for silicon oxide particles, the lithiation state must be considered when determining the particle radius due to the huge volume changes of the particles during lithiation of about 118% for silicon oxide and 300% for pure silicon.⁵ In general, the mean particle radii for the negative electrodes are bigger compared to those of the positive. The mean particle radii of the M50 and M50T are almost identical with

$5.22 \mu\text{m}$ and $5.14 \mu\text{m}$, respectively. The mean particle radius for the positive electrode of the LG M50LT is smaller with $4.50 \mu\text{m}$.

Chemical analysis.—The battery cells were further examined in terms of their elemental composition. For this, EDS and GDOES measurements were performed as outlined in the following sections.

EDS measurements.—By using EDS, the elemental composition of the electrode particles were visualized and the element composition of the electrodes on a microscopic scale determined. Figure 9 illustrates the EDS results for the negative electrodes of the M50T

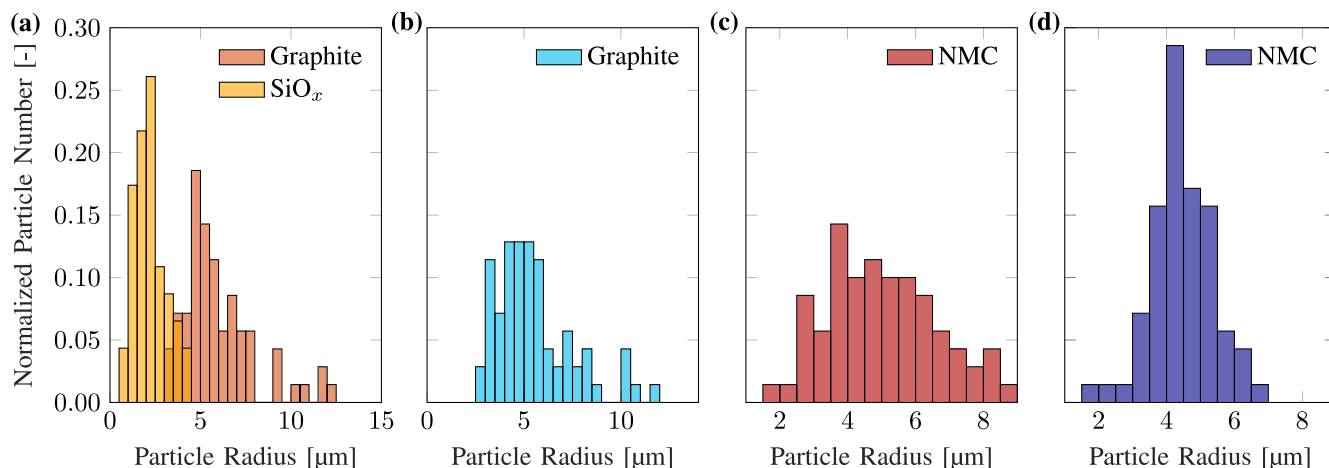


Figure 8. Particle size distributions based on the SEM images in Fig. 5 for the negative electrodes of the LG M50T (separated for graphite and SiO_x) (a) and the M50LT (b) as well as for the positive electrodes of the M50T (c) and M50LT (d).

(a) and M50LT (b) with a field of view of $128 \times 96 \mu\text{m}$ and connects the SEM images with the element composition. The red particles refer to graphite and the yellow particles to silicon oxide (SiO_x). At this point it should be emphasized that EDS can only determine the elemental composition and that the silicon oxide particles represent an overlap of detected silicon and oxygen elements. It is striking that no silicon elements were detected for the M50LT whereas silicon oxide particles are visible for the M50T. Although the EDS measurements were conducted with several samples and at different location, silicon or silicon oxide was not found for the M50LT. This implies that the negative electrode of the M50LT only consists of graphite as an active material, whereas the M50T is made of a composite electrode comprising graphite and silicon oxide (SiO_x). The silicon oxide particles are approximately homogeneously distributed and are generally smaller than the graphite particles as outlined in the previous subsection. For all negative electrodes, sulfur was detected in the background. Since sulfur is known to be a natural pollution of graphite and for better visualization, sulfur was not displayed in the images below.⁵⁸ Furthermore, phosphor and fluorine were detected in the electrode which can be interpreted as part of the SEI and are therefore also not displayed in the EDS images.^{59,60} Figure S4 in the supplementary information shows that silicon and oxygen can be found at the same electrode locations which implies that silicon oxide is used besides graphite as an active material of the M50T. Since only the surface of the particles is detected with EDS, their inner composition could differ from that of the surface (e. g. pure silicon could be present inside the particles). Oxygen was not only detected in correspondence with silicon but

also across the whole electrode. However, the intensity was significantly lower in these remaining regions and can thus be attributed to the SEI.⁶¹ In addition, copper was detected with a low intensity, which can likely be attributed to the current collector. However, contributions from current collector dissolution and deposition may also be present and cannot be fully excluded at this stage. Besides these elements, no others were found with EDS.

The EDS data of the positive electrodes of the M50T and M50LT indicate the presence of nickel, manganese, cobalt, oxygen, carbon and aluminum. Since the positive electrodes consist of a lithium metal oxide where the single elements are combined in a lattice structure, EDS only shows the overlay of these elements and cannot assign the elements to single particles as shown for the composite negative electrode. The particle-specific element composition is presented in the supplementary data in Fig. S5. Furthermore, the EDS data illustrates that carbon surrounds the NMC particles in the shape of carbon fibers in the nanometer range. Sulfur was also detected in the positive electrodes in regions where carbon is present. This underlines the assumption that the presence of sulfur is due to pollutions in graphite.⁵⁸

To determine the element composition of the electrodes on a microscopic scale, EDS was conducted with a wider field of view of $674 \times 505 \mu\text{m}$. The measurements were performed three times and the averages calculated. The element composition determined with EDS is summarized in Table II. The mass percentage of silicon in the negative electrode of the M50T was determined to 2.35 mass-%. The silicon content for the LG M50T was simulatively determined by Ai et al. (2022)¹⁶ to 2 volume-%. Assuming that the density of

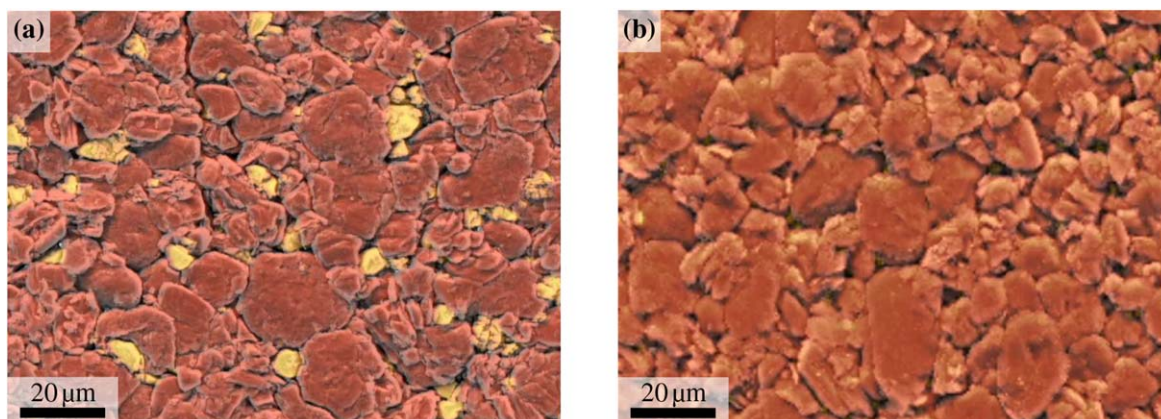


Figure 9. EDS measurements of the negative electrode of the M50T (a) and M50LT (b) with a field of view of $128 \times 96 \mu\text{m}$. The red particles illustrate graphite and the yellow particles silicon oxide.

Table II. Chemical composition of the negative and positive electrodes of the LG M50T and M50LT determined with EDS and GDOES.

Element	Unit	Negative electrode			
		LG M50T		LG M50LT	
		EDS	GDOES	EDS	GDOES
Carbon	mass-%	84.97	89.38	86.82	89.32
Silicon	mass-%	2.35	2.37	0.00	0.00
Oxygen	mass-%	12.47	6.07	12.42	7.76
Copper	mass-%	0.21	0.81	0.77	1.66
Lithium	mass-%	not detectable	1.38	not detectable	1.25

Element	Unit	Positive electrode			
		LG M50T		LG M50LT	
		EDS	GDOES	EDS	GDOES
Nickel	mol-%	24.71	17.42	24.49	22.22
Manganese	mol-%	1.90	1.26	1.49	1.14
Cobalt	mol-%	3.52	3.02	2.39	2.39
Aluminum	mol-%	0.23	0.04	0.24	0.18
Oxygen	mol-%	52.33	71.36	49.00	66.38
Carbon	mol-%	17.30	0.96	22.39	1.38
Lithium	mol-%	not detectable	5.94	not detectable	6.31

silicon is similar to that of graphite this matches with the mass fraction measured by EDS in this study. Chen et al. (2020)¹⁵ used EDS as well to evaluate the chemical composition of the M50 and measured the silicon oxide content to 10 volume-%. They also determined the stoichiometry x in SiO_x to 0.64 by using EDS. This leads to a silicon mass content of 7.3 mass-%. In the present work, the sum of the silicon and oxygen content was evaluated to 14.82 mass-%. The oxygen content measured with EDS is the total oxygen content in the sample and does not only belong to silicon oxide. Although the samples have been washed and dried before the measurements and consequently electrolyte residues should be removed almost completely, the composition of the SEI influences the oxygen content in the electrodes. Especially, since the SEI is present on the surface of the electrodes and has not been removed before, their chemical compositions were measured superimposed with the electrode composition by EDS. Hence, taking the measured oxygen content and connecting it with silicon to silicon oxide might lead to misinterpretation. Consequently, EDS data only cannot be used to calculate the stoichiometry x of oxygen in SiO_x . Thus, the composition of the negative electrodes of the M50 and M50T need to be analyzed further which will be done in the following sections. The carbon content of the M50T in the negative electrode was determined to 84.97 mass-%.

The composition of the negative electrode of the LG M50LT revealed a carbon and oxygen content of 86.82 mass-% and 12.42 mass-%, respectively. Silicon was not detected at all for the LG M50LT. This underlines that the oxygen content measured with EDS is not proportional to the silicon content. The copper content was detected to be below 1% for both electrodes. Due to the porous structure of the electrodes, the electron beam of the SEM/EDS penetrated through the electrode coating layers and detected the composition of the underlying current collector although the surfaces of the electrodes were analyzed. This explains the copper content measured.

In addition, the chemical composition of the positive electrodes were also determined on a micro-scale level using EDS with a field of view of $674 \times 505 \mu\text{m}$. For the positive electrode of the M50T, the molar ratio of nickel, manganese, cobalt and aluminum was detected to 7.02:0.54:1:0.065; for the M50LT to 10.27:0.63:1:0.1. Thus, the electrodes can approximately be classified as NMC-811. The aluminum content is attributed to the current collectors analogously to copper in the negative electrodes. The molar ratio of nickel to oxygen is 1:2.12 for the M50T and 1:2 for the M50LT.

Besides small quantities of fluorine and phosphor below 0.1 mol-% no other elements were detected. These quantities are assigned to the cathode electrolyte interphase (CEI).⁶² Due to the very low energy X-rays for the Li-K α radiation of 54 eV,⁶³ lithium is not detectable with the EDS detector used.

Chen et al. (2020)¹⁵ used ICP-OES and EDS data to determine the stoichiometric composition of the positive electrode of the M50 to $\text{Li}_{0.833}\text{Ni}_{0.80}\text{Mn}_{0.08}\text{Co}_{0.08}\text{Al}_{0.04}\text{O}_2$. Compared to the values determined with EDS in the present study, the manganese molar fraction is about half of the molar fraction of cobalt for both the M50T and M50LT and not identical as determined by Chen et al. Further, the aluminum fraction is one ten of the cobalt fraction for the M50T and M50LT but half for the M50. The ratio of nickel to cobalt is nearly identical for the M50 and M50T. Nickel is detected to be higher for the LG M50LT compared to the other cells. According to Ref. 64 a higher nickel content leads to a higher specific capacity but also the thermal instability increases. Manganese is inserted into the lattice structure to overcome these instabilities⁶⁵ and cobalt helps to enhance the electrochemical performance and cycle stability, although cobalt is known to be an expensive and scarce metal.^{66,67}

GDOES measurements.—In contrast to EDS which measures the chemical composition on a microscopic scale, GDOES is used to determine the composition of the electrodes on a macroscopic level averaged on a sputtering crater of 2.5 mm in diameter. The measured electrode compositions are visualized over the electrode thickness in Fig. 10.

After the transient response at the beginning of the measurements, the chemical composition of the electrodes is nearly constant. For the negative electrodes, the copper content rises after 86 μm and 74 μm for the M50T and LT (Figs. 10a and 10b), respectively, since the electrode materials are sputtered through at this point followed by the current collector made of copper. Copper is also detected across the electrode. Due to the porous structure of the electrode, besides the active material and additives, copper is constantly sputtered away from the current collector and therefore detected with GDOES and EDS. The positive electrode (Figs. 10c and 10d) behaves analogously and shows a rise in aluminum content after having sputtered through the electrode.

The chemical composition was then averaged over the electrode thickness for the region after the transient process and before sputtering the current collector where the element composition is nearly constant. The results are summarized in Table II. The

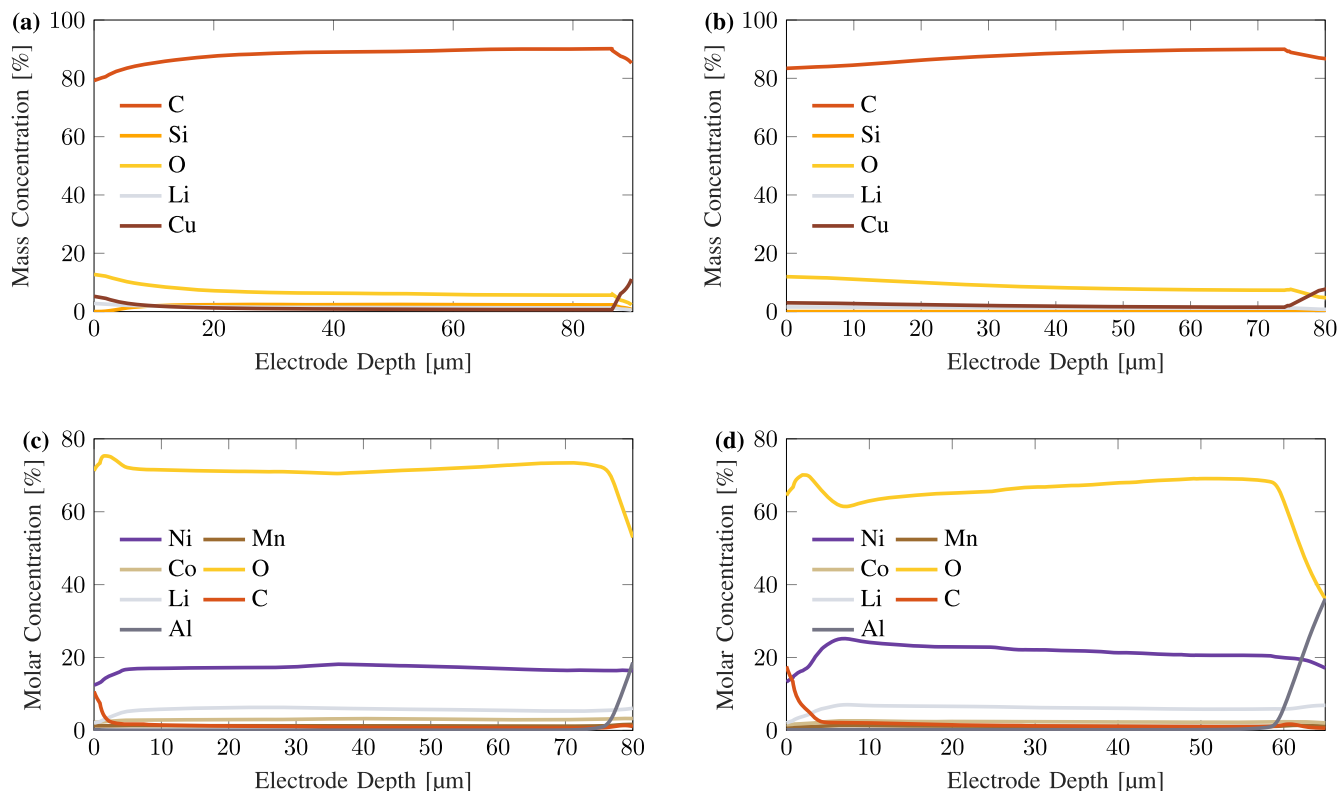


Figure 10. Mass composition over the electrode depth of the negative electrode of the M50T (a) and M50LT (b) and molar composition of the positive electrode of the M50T (c) and M50LT (d) determined with GDOES.

negative electrode mainly consists of carbon (>84 mass-%) for both the M50T and LT. A silicon content of 2.37 mass-% was detected for the M50T which is almost identical to what was detected by EDS (2.35 mass-%). The oxygen content was measured to be about half of the value measured by EDS. While EDS analyzes the surface of the electrodes rather than the inner, the electrode material is sputtered away with GDOES until the current collector is reached and hence the chemical composition inside the electrode material is characterized. Since the SEI is present on the surface of the electrode, EDS also detects the chemical composition of the SEI. The higher oxygen content on the surface detected with EDS compared to inner part of the electrode analyzed by GDOES can thus be interpreted as the results of the SEI. The oxygen fraction was measured to be lower for the M50T compared to the LT although the M50T contains silicon oxide. Further, small quantities of copper were detected below 2 mass-%. The reason for this is as described in the previous section. Lastly, the lithium molar ratio between carbon and lithium is 37.6:1 and 41.2:1 for the M50T and LT, respectively. Since the electrode samples were investigated after discharging a full-cell to 2.5 V, the negative electrode has been almost fully delithiated (as will be outlined in the Section Electrode Balancing). Thus, the detected lithium contents for the negative electrodes are very low.

For the positive electrode, the ratio of nickel, manganese and cobalt determined with GDOES is 5.77:0.42:1 for the M50T and 9.3:0.48:1 for the M50LT. Therefore, the cells can also be classified as NMC-811 with GDOES. As determined with EDS, the content of nickel in the M50LT is higher compared to the M50T. The molar ratio of manganese to cobalt is nearly identical for the two cells and is approximately 1:2 as determined with EDS as well. The oxygen and carbon contents differ between the EDS and GDOES measurement: The carbon content was determined to be about one order of magnitude higher with EDS compared to GDOES. The sums of carbon and oxygen contents for both measurements techniques, however, are almost the same. Due to chemical side reactions with the electrolyte during formation, the chemical composition of the

CEI present at the surface of the electrode particles is analyzed with EDS as well which leads to higher carbon contents for the EDS measurement.^{62,68} The molar ratio of nickel to oxygen is 1:4.1 and 1:2.99 for the M50T and LT, respectively. Contrary to EDS, GDOES can detect the lithium content. Lithium and nickel are detected in a ratio of 1:2.93 and 1:3.52 for the M50T and LT, respectively.

To summarize the chemical analysis, the ratios of graphite and silicon are detected similarly with EDS and GDOES for the negative electrodes. The silicon content is determined at an average of 2.36 mass-% $\pm$ 0.01%. For the positive electrode, the ratios of nickel, manganese and cobalt are identified almost identical with EDS and GDOES. The nickel content is evaluated to be higher for the M50LT compared to the M50T for both EDS and GDOES. Only the carbon and oxygen contents probably differ due to the CEI.

Electrochemical analysis.—Experimental cells in half-cell configurations (commercial electrode versus lithium metal) with a lithium reference ring were built up and investigated to determine their pOCPs. DVA, GITT and EIS measurements were also conducted on experimental cell level. Further, the cylindrical cells were used to analyze the DVA curve and to calculate the electrode balancing. The results of these measurements are discussed subsequently.

Open circuit potential and voltage.—The half-cell pOCPs of the negative electrodes of the M50, M50T and M50LT are presented in Figs. 11a and 11b separated for lithiation and delithiation, respectively. The half-cell pOCPs of the LG M50 were extracted from Ref. 15. By comparing the lithiation and delithiation process of the electrodes, a significant hysteresis characterizes the behavior, especially for the M50 and M50T. An overlay of the lithiation and delithiation pOCPs for the negative electrodes can be found in Fig. S8 in the supplementary information. The pOCPs of the M50 and M50T are nearly identical whereas the pOCP of the M50LT differs from the others. This behavior results from the different active

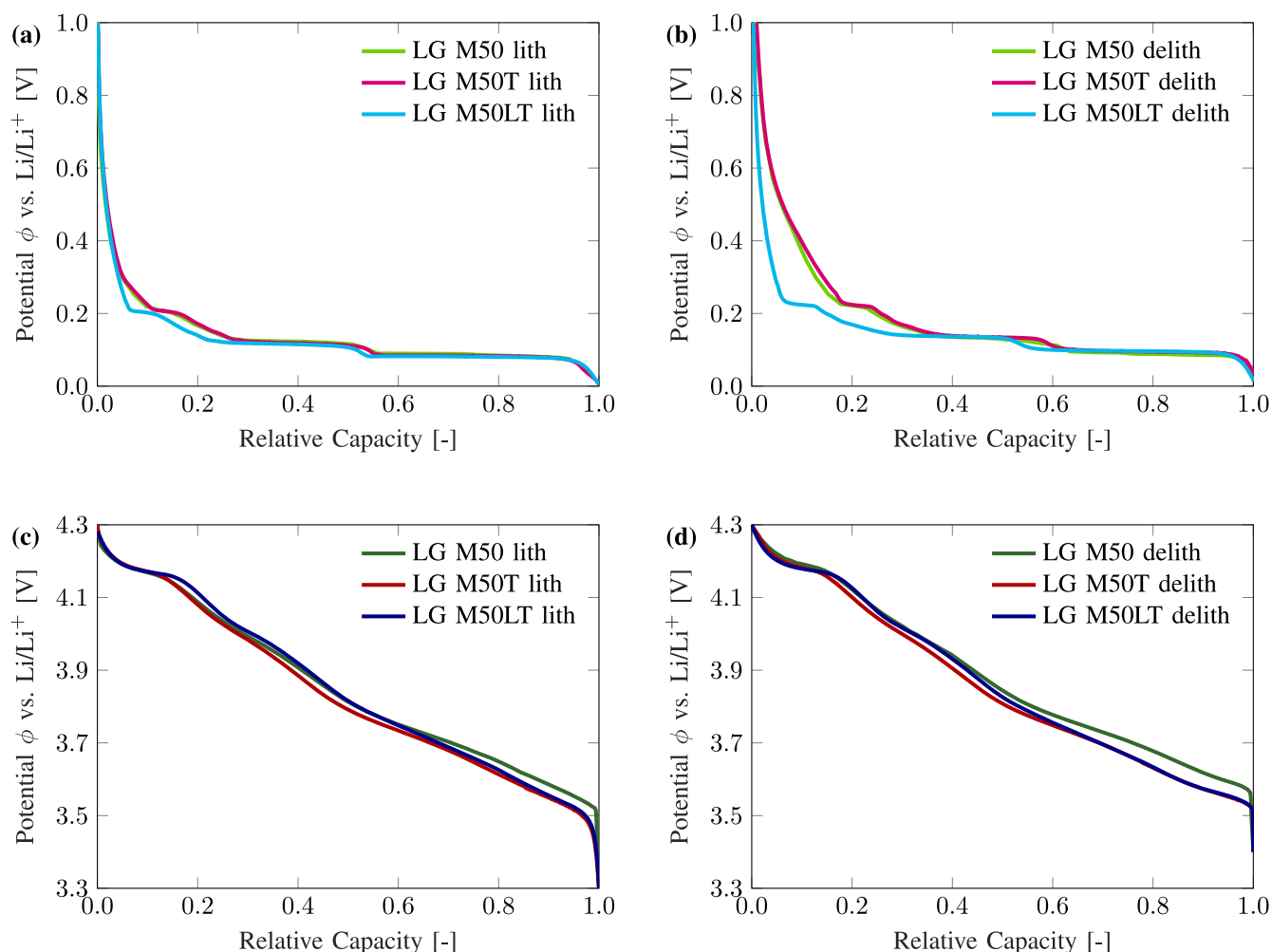


Figure 11. Half-cell pOCPs of the negative electrodes of the M50, M50T and M50LT for lithiation (a) and delithiation (b) and for the positive electrodes for lithiation (c) and delithiation (d). Data for the LG M50 was extracted from Ref. 15.

materials present in the negative electrodes as outlined in the chemical analysis. The M50LT only consists of a graphite electrode with a small hysteresis.⁴³ In contrast, the other cells contain a composite negative electrode made of graphite and silicon oxide. Due to the silicon content, the pOCP changes and the hysteresis becomes more significant.⁵ The gradual course of the pOCP of the negative electrodes can be attributed to the graphite active material due to the layered-wise intercalation of lithium into the graphite layered structure as further outlined in Ref. 43.

The half-cell pOCPs of the positive electrodes are displayed in Figs. 11c and 11d for lithiation and delithiation, respectively. The hysteresis is less significant compared to the negative electrode. For lithiation and delithiation, the initial slope at low relative capacities is identical for the three cells. In the middle of the curve, the M50LT has a higher potential compared to the M50T. This may be due to the different nickel contents in the electrodes. The M50 positive electrode is characterized by a less steep gradient at high relative capacities but generally follows the trend of the other cells. The potential curve of the positive electrode also implies that the active material is NMC-811. The characteristic flat voltage curve at high voltages (4.0–4.2 V) was also observed in other publications.⁶⁹

The area-specific, volume-specific, and mass-specific capacities were calculated based on the coating area in the half-cell setup (circular coin with 18 mm in diameter) and the capacity obtained during the pOCP experiments. The results are summarized in Table I. Since the electrode thickness differs between the cells, the area-specific capacity is inappropriate for comparison. Due to the

low electrode thicknesses of the M50LT, its area-specific capacities are the lowest. Considering the electrode thicknesses, the volume-specific capacities are calculated. The volume-specific capacities of the negative and positive electrode of the M50 achieved in the experimental setup were $595.1 \text{ mAh cm}^{-3}$ and $609.4 \text{ mAh cm}^{-3}$, respectively.¹⁵ For the experimental cells of the negative and positive electrode of the M50T, capacities of $601.4 \text{ mAh cm}^{-3}$ and $670.9 \text{ mAh cm}^{-3}$ were recorded and those of the M50LT were determined to $578.3 \text{ mAh cm}^{-3}$ and $719.8 \text{ mAh cm}^{-3}$ for the negative and positive electrodes, respectively. With regard to the negative electrode, the M50 and M50T show similar volume-specific capacities of approximately 600 mAh cm^{-3} which is about 4% higher than that of the M50LT. The lower volume-specific capacity of the negative electrode of the M50LT compared to the M50T and M50LT is caused by the chemical composition of the electrodes, i.e. by the absence of silicon in the M50LT. The higher volume-specific capacity of the positive electrode of the M50LT can be attributed to the higher nickel content compared to the M50 and M50T as outlined in the chemical analysis. The M50 also shows a lower volume-specific capacity compared to the M50T which might be caused by the higher current applied in Ref. 15 of C/50 (C/100 was used in this study) which increases the overpotentials especially in experimental setups. The voltage limits used in both studies were comparable: 0.005 V to 1.5 V for the negative electrode of the M50 in Ref. 15 and 0.005 V to 1.3 V for the M50T and LT; 2.5 V to 4.2 V for the M50 positive electrode in Ref. 15 and 3.3 V to 4.3 V for the M50T and LT. Due to the high gradient in pOCP below voltages of

3.3 V for the positive electrode and above 1.3 V for the negative electrode, the differences in the lithiation state between the voltage range of Chen et al. and that used in this study can be neglected.

Multiplying the determined volume-specific capacity with the total electrode volume of the cylindrical battery calculated in the geometrical analysis results in the capacity storable in each electrode considering the voltage limits of the half-cells. Hence, the negative electrode capacity is 5727mAh for the M50T and 5397mAh for the M50LT, although the total volume of the negative electrodes are almost identical. The increase in capacity for the M50T can be attributed to silicon oxide. Due to the high specific capacity fraction, a small amount of silicon added to the graphite active material leads to a significant increase in capacity.

On the positive electrode side, the capacity calculated with the volume-specific capacity and the total electrode volume is 5412mAh for the M50T and 5296mAh for the M50LT. The calculated capacity only varies by 2.14% for the positive electrodes, albeit the electrode volume of the M50LT is by 9.6% smaller than that of the M50T. This can be explained because of the higher nickel content and thus the higher specific capacity in the positive electrode of the M50LT as discussed in the chemical analysis.

For the M50T, the negative electrode shows a 5.8% higher capacity than the positive electrode. This capacity retention can probably compensate the high capacity loss of the silicon oxide particles during degradation mainly as a result of particle cracking due to the high volume changes of silicon.^{22,23} For the M50LT, the capacities for the negative and positive electrode are almost identical (negative electrode has 1.9% more capacity). Since pure graphite electrodes generally degrade less rapidly compared to silicon-containing electrodes,⁷⁰ the capacity retention does not need to be as high as for the M50T. The constructive capacity of the positive electrode should be higher than that of the negative electrode capacity since lithium from the positive electrode is utilized irreversibly to form the SEI on the negative electrode side during formation. After the formation procedure, the remaining negative electrode capacity must be higher to reduce the probability of lithium plating. All three cells under investigation show a higher available capacity for the negative electrode than for the positive to lower the aging rate and prevent lithium plating.⁷¹

The mass-specific capacity was computed by the volume-specific capacity and the electrode coating density ρ_{coating} . Since the electrode coating densities of the positive electrodes are similar for all three cells, the same trend can be observed for the mass-specific capacity as for the volume-specific capacity. The negative electrode of the M50T has the highest mass-specific capacity which is higher than that of the M50LT due to the use of silicon. The low mass-specific capacity of the M50 may result from the comparatively high electrode coating density computed by Chen et al. (2020).¹⁵

Since the electrode operating window was exceeded for the half-cell measurements compared to the cylindrical cell (as will be outlined in the following Section Electrode Balancing), the capacities determined based on the half-cells are slightly higher than the capacity of the cylindrical battery cells mentioned in the data sheet.

Electrode balancing.—To ascertain the balancing and thus operating window of the electrodes, the electrode balancing is simulatively investigated for the pOCV of the cylindrical batteries by applying (10). Pristine batteries were used for this investigation, i.e. the formation procedure was done by the manufacturer but the cells were not cycled after being delivered. For aged cells, the operating window of the electrodes change due to non-symmetrical degradation of the electrodes as discussed in Ref. 31. Figure 12 visualizes the results for the M50T (a) and LT (b) during discharge. The black dashed lines mark the voltage limits for the full-cell and the gray dashed lines indicate the voltage limits of the half-cells. Since the difference between LG M50T and LT was more significant for discharging, the balancing was not considered for charging.

For the pristine M50T cylindrical cell cycled from 2.5 V to 4.2 V, the negative electrode operates in a potential window of 1.099 V to

0.092 V and the positive electrode in a window of 3.554 V to 4.278 V. The operating windows for the LG M50LT for the negative and positive electrode are 1.096 V to 0.092 V and 3.550 V to 4.282 V, respectively. The operating windows of the negative electrodes are nearly identical and only the upper voltage limit is higher for the M50T. For the positive electrodes, the voltage limits for the M50LT are increased, i.e. the upper voltage limit is increased by 4 mV and the lower voltage limit decreased by 4 mV. Consequently, the positive electrode of the M50LT is more lithiated and delithiated. The RMSE between measured and simulated pOCV for the M50T during discharge is 5.6 mV and for the M50LT it is 3.7 mV. The voltage difference is the largest in the area of low full-cell SoCs where the gradients in the negative electrode pOCVs are the highest.

The scaling factor α_{neg} is 1.0082 and the shifting factor $\beta_{\text{neg}} = -0.65\%$ for the negative electrode of the M50T. α_{pos} and β_{pos} for the M50T are 1.070 and -11.45% , respectively. For the M50LT, the scaling factors of the negative and positive electrode are $\alpha_{\text{neg}} = 1.0153$ and $\alpha_{\text{pos}} = 1.049$ and the shifting factors are $\beta_{\text{neg}} = -0.22\%$ and $\beta_{\text{pos}} = -9.65\%$, respectively. The differences in the scaling and shifting factors between the M50T and LT also show that the positive electrode of the M50LT is less stretched ($\alpha_{\text{pos,M50T}} > \alpha_{\text{pos,M50LT}}$) and less left-shifted ($\beta_{\text{pos,M50T}} > \beta_{\text{pos,M50LT}}$). Therefore, a wider operating range is used for the M50LT positive electrode. The negative electrode of the M50T operates in nearly the same window as the M50LT but the M50LT is slightly more stretched and right-shifted meaning that the operating window is minimally reduced.

Consequently, the negative electrode of the M50T is cycled in almost the same voltage window as the M50LT. However, the capacity window of the positive electrode of the M50LT is increased. Furthermore, due to the higher nickel content in the M50LT, the energy density of the positive electrode of the M50LT is higher compared to that of the M50T. By combining both effects, the space required to deliver the same capacity for the positive electrode is reduced and more space is available for the negative electrode in the 21700 cell can. The lower total electrode volume for the positive and the higher total electrode volume for the M50LT in comparison with the M50T can be seen in the geometrical analysis in Fig. S3.

The capacity reduction on the negative electrode side due to the absence of silicon has thus been compensated by a smaller but more energy dense positive electrode. Hence, the positive electrode volume could be decrease and a higher total electrode volume on the negative electrode side could be implemented by keeping the total capacity of the cylindrical cell nearly constant. On cell level, the capacity difference between the M50T and M50LT is only 1.6% according to the data sheet although no silicon was used on the negative electrode side of the M50LT. In addition, the nominal voltage for the M50LT is specified higher in the data sheet (3.69 V) compared to the M50 and M50T (3.63 V) which might correlate to the higher nickel content and the balancing shift of the positive electrode compared to the M50 and M50T.

Since more electrode coating volume was required to compensate the absence of silicon in the negative electrode of the M50LT, increasing the thickness of the negative electrode only and keeping the lateral dimensions equivalent to the M50 or M50T could limit the diffusion processes through the thickness of the negative electrode. Increasing the electrode length and simultaneously decreasing the thickness of the negative electrode could help to overcome the diffusion limits. Hence, also the positive electrode needs to be enlarged. This could explain why the electrodes are longer and thinner in case of the M50LT compared to the M50 and M50T.

Simulation of the composite OCP.—Besides determining the chemical composition of the negative electrode of the M50T with EDS and GDOES, the active silicon content of the composite electrode can also be calculated by combining the OCPs of pure graphite and pure silicon appropriately to fit the composite OCP. By

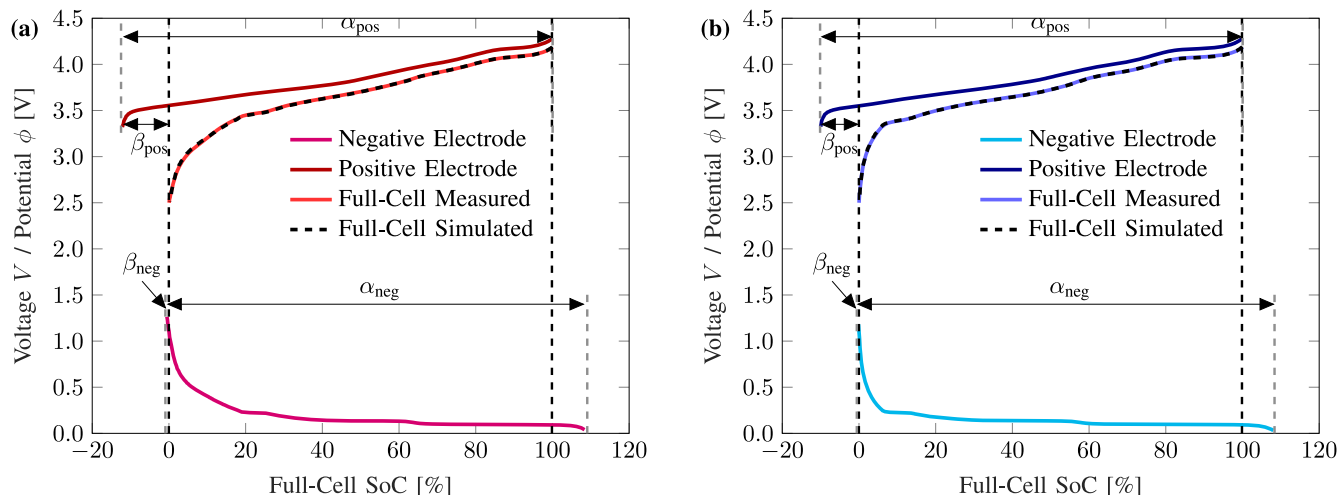


Figure 12. Simulated electrode balancing of the negative and positive electrodes of the M50T (a) and M50LT (b) during discharge compared with the measured full-cell cylindrical battery pOCV.

minimizing the RMSE between simulation and measurement, a silicon capacity fraction of 13.1% was found to match the measured data best. Consequently, 86.9% of the capacity in the composite electrode refers to graphite. Kirkaldy et al. (2022) also reconstructed the composite OCP from pure silicon and graphite and found a capacity fraction of silicon of 15% which is in the same range as calculated in this study.²² The simulation results are illustrated in Fig. 13 showing the used graphite and silicon OCPs separately for lithiation and delithiation. The significant hysteresis of silicon can also be seen here which leads to a hysteresis behavior of the composite electrode as well. The RMSE of the simulated and measured lithiation and delithiation processes are 6.0 mV and 9.2 mV, respectively. The absolute potential difference between simulated and measured potential curve for lithiation and delithiation are depicted in the supplementary information in Fig. S9. The largest potential errors occur at low lithiation states where the potential gradient is high since the data for the graphite phase was extrapolated for potentials above 0.4 V using⁴³ and⁴⁴.

During the alloy formation of Si to $\text{Li}_{15}\text{Si}_4$ a theoretical specific capacity of $C_{\text{Si,spec}} = 3579 \text{ mAh g}^{-1}$ can be achieved.¹ Due to the balancing and the resulting reduction of usable stoichiometry in the negative electrode of commercial cylindrical battery cells, a smaller amount of capacity storage can be used compared to the theoretical value. This phenomena can be seen in Section Electrode Balancing. When the cell is cycled between 2.5 V and 4.2 V, the negative electrode potential is within the range of 1.099 V and 0.092 V. Since $\text{Li}_{15}\text{Si}_4$ is the highest lithiation state at a silicon potential of 0.05 V⁷²

only 81.0% of the stoichiometry of $\text{Li}_{15}\text{Si}_4$ is used during lithiation in full-cell operation (see Fig. 13) because of the balancing. In addition, the M50T contains silicon oxide (SiO_x) and not pure silicon as an active material. As outlined in Ref. 10, 73 and, 74 silicon oxides (SiO and SiO_2) undergo irreversible reactions with lithium to lithium oxide (Li_2O) and lithium silicates (e.g. Li_4SiO_4 , $\text{Li}_2\text{Si}_2\text{O}_3$ and $\text{Li}_2\text{Si}_2\text{O}_5$) during the first formation cycles. Only the chemical reactions to Li_ySi_z alloys are reversible.¹⁶ Although these irreversible reaction products act as a buffer to reduce the volume expansion, the amount of cyclable silicon is also lowered. Assuming that 75% of the silicon content and therefore 75% of the theoretical specific capacity is used in the cylindrical cells, the usable specific capacity is $C_{\text{Si,spec,usable}} = 2684.3 \text{ mAh g}^{-1}$. The value of the theoretical capacity of silicon oxide is also in the same range calculated by Ref. 9 and 10.

Using the simulated capacity fractions of silicon and graphite and applying them in case of the cylindrical battery M50T with a capacity of 5000mAh, the amount of capacity delivered by silicon C_{Si} and graphite C_{Gr} are therefore 655mAh and 4345mAh, respectively. Dividing the capacity C_{Si} by the usable theoretical specific capacity of silicon $C_{\text{Si,spec,usable}}$, the mass of silicon m_{Si} in the cylindrical cell is calculated to 0.244 g. Using the electrode loading $\rho_{\text{A,neg}}$ and total coating area of the negative electrode $A_{\text{coating,neg}}$ of the M50T from the geometrical analysis, the total mass of the coating of the negative electrode $m_{\text{coating,neg}}$ is 11.02 g. This leads to a mass fraction of silicon of $\epsilon_{\text{Si}} = 2.21\%$. This value was also approximately measured with EDS and GDOES.

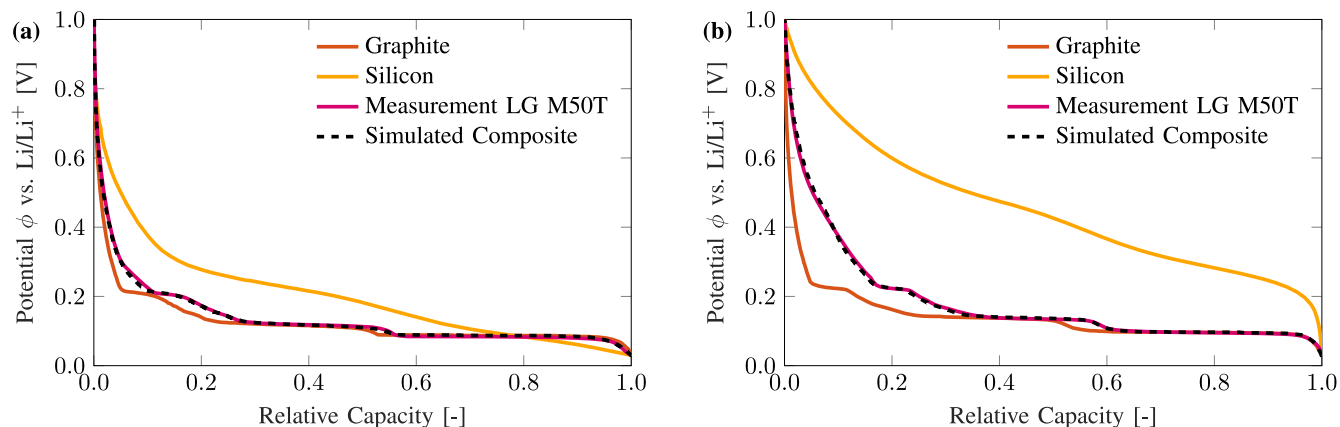
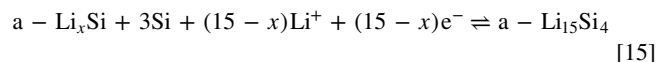
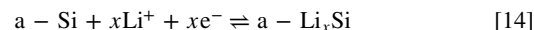


Figure 13. Measured and simulated half-cell OCP for the negative electrode of the LG M50T for lithiation (a) and delithiation (b) based on the OCPs for pure graphite and pure silicon from Ref. 43, 44, and 45.

Differential voltage analysis.—To investigate phase changes of the active materials during (de-)lithiation and to determine the silicon content as outlined in Ref. 33 the differential voltage was analyzed. Here, the pOCVs of the cylindrical cells M50T and M50LT for charge and discharge (Fig. 14a) were used to determine the derivation dV/dQ^{-1} (Fig. 14c). The DVAs of the pOCPs for the negative and positive electrodes can be seen in Figs. 14b and 14d, respectively. The capacity axis for all plots refer to the voltage curve of the full-cell from Fig. 14a. The appropriate half-cell pOCPs are taken from Fig. 12. The specific capacities of the half-cells were calculated using the electrode loading ρ_A , the area of the electrodes in the experimental setup and the capacity determined in the half-cell setups. The separate DVAs for the half-cell pOCPs of the negative and positive electrodes help to interpret the peaks in the full-cell DVA by distinguishing between phase changes in the negative or positive electrodes.

There are two separate peaks striking in the pOCV and the pOCP of the negative electrode of the M50T in low capacity ranges. These two peaks correspond to the reversible phase transformation reactions of silicon occurring while silicon is lithiated above 0.05 V

versus lithium. As shown in the Section Electrode Balancing, the negative electrode operates above 0.05 V on a full-cell level within the voltage limits given by the manufacturer—at least at begin of life. The homogeneous two-phase transformation was confirmed by Wang et al.⁷² in an in-situ transmission electron microscope (TEM) experiment. The reactions happening are presented in the following reaction equations.⁷⁷



In the first reaction, amorphous silicon (a-Si) reacts to amorphous $\text{Li}_{x\text{Si}}$ (a- Li_xSi) with x considered to be below 2.^{45,54} Lithiating silicon more leads to the final amorphous product a- $\text{Li}_{15}\text{Si}_4$. The reaction equations can be assigned to the peaks labeled as Si_1 and Si_2 .

The peaks attributed to graphite occur due to the stepwise intercalation of lithium in the graphite layered structures as discussed in Ref. 31, 43 and 78 referring to the phase diagram of

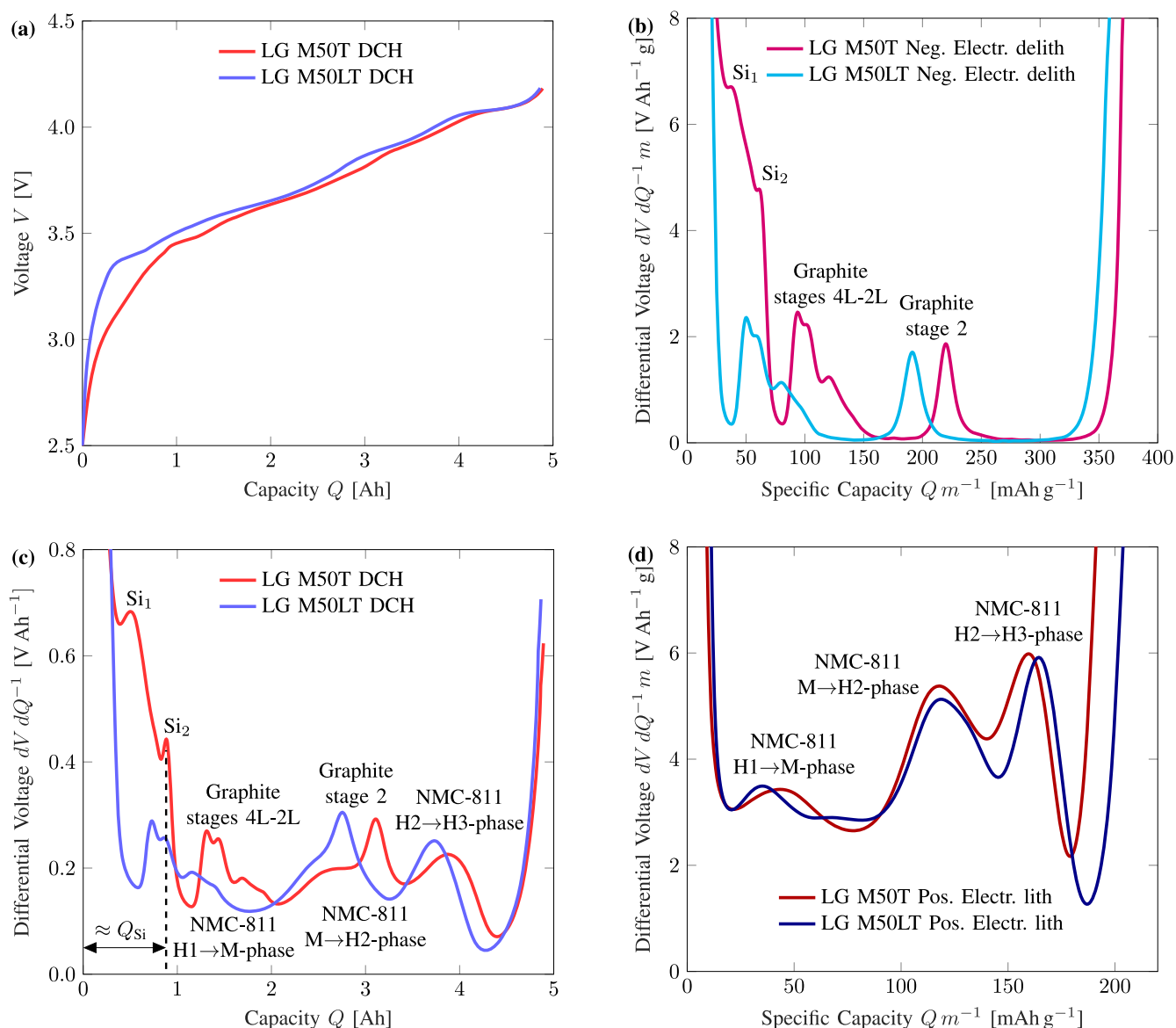


Figure 14. Voltage curve (a) and differential voltage analysis of the M50T and M50LT cylindrical cell over the capacity (c) for discharge. Differential voltage analysis of the negative (b) and positive (d) electrodes of the M50T and M50LT over the mass-specific capacity for delithiation and lithiation, respectively. Labeling adopted from Ref. 31, 43, 75, and 76.

graphite. The name of the stages result from the order of lithium ions in every n -th interlayer. The letter L is added to the stage name if the lithium ion ordering is liquid-like and disordered. For instance, stage 2L indicates that every second interlayer is filled with lithium ions with a disordered in-plane arrangement. In contrast, the lithium ions in stage 2 are ordered so that every second layer is filled with LiC_6 .⁴³ Due to silicon, the specific capacity of the negative electrode material is higher for the M50T compared to the M50LT. For the M50LT, only the phase transitions of graphite can be seen in the half-cell DVA concluding that there is no silicon present in the M50LT.

As discussed in the Sections before, the positive electrodes of both the M50T and M50LT can be classified as NMC-811 which leads to similar potential and DVA curves (see Fig. 14d). Nevertheless, the specific capacity is higher for the positive electrode of the M50LT due to the higher nickel content,^{64,79} as determined with both EDS and GDOES (see Section chemical analysis). In the positive electrode, NMC-811 undergoes three phase changes during (de-)lithiation as marked in Fig. 14d and described in Ref. 76. The phase transition at low full-cell voltages and low positive electrode potentials, i.e. high lithiation states, results from the transition from a hexagonal (H1) to a monoclinic (M) lattice structure. The local maximum and minimum at these high lithiation states of the positive electrode of the M50LT are less prominent compared to the M50T which can also be attributed to the different chemical compositions. The next peaks in medium full-cell voltage regions occur due to the phase transition from the M to the H2 phase and in high voltage regions, the peaks belong to the phase transition from the H2 to H3 phase of NMC. The latter two mentioned phase transition were only reported for Ni-rich NMC materials such as NMC-811.⁷⁶ Therefore, the DVA also reveals that the positive electrodes can be classified as NMC-811.

The peaks mentioned in the half-cell negative and positive electrode DVAs also appear in the full-cell DVA of the cylindrical cell. By superimposing the half-cell DVAs appropriately, the full-cell DVA can be created.

For the half-cell negative electrode of the M50T, a silicon capacity fraction of 16.5% was determined from Fig. 14b based on the capacity at peak Si_2 . The full-cell shows a capacity fraction of silicon of 18.1% as marked in Fig. 14c. In comparison, the simulated capacity fraction of silicon to fit the OCP of the negative electrode (see Section Simulation of the Composite OCP) was calculated to 13.1% which corresponds to a difference of 3.4% compared to the half-cell DVA. With regard to the full-cell DVA, the difference to the simulated capacity fraction is 5.0%. Using the assumptions outlined in Section Simulation of the Composite OCP, the mass fraction of silicon can be evaluated from the capacity fraction. This leads to an active mass fraction of silicon in the negative electrode of the M50T of 2.79% and 3.06% using the half-cell and full-cell DVAs, respectively. Due to the potential curve of silicon, silicon is more active in lower full-cell voltage regions but graphite also participates in the reactions occurring on the negative composite electrode side. Therefore, the capacity fraction of silicon can be overestimated by using the DVA which is shown here as well. Since the negative electrode is less lithiated in the full-cell compared to the half-cell (see Section Electrode Balancing), graphite is less cycled relative to silicon and the determined capacity fraction of silicon is higher in case of the full-cell DVA. This explains the discrepancy between half-cell and full-cell DVA. The values for the silicon mass fraction are in the same range as measured by EDS and GDOES. The DVA for the charging process and the ICA can be found in the supplementary information in Figs. S10–S12.

To summarize, the silicon mass fraction in the negative electrode was determined using different measurement techniques: EDS, GDOES, simulated composite OCP and DVAs on half-cell and full-cell level. Using the chemical analysis method EDS and GDOES, the silicon mass fraction was determined to 2.35% and 2.37%, respectively. The simulated composite OCP delivered a capacity fraction of 13.1% leading to a mass fraction of 2.21%.

Finally, the mass fraction of silicon determined using the DVA on half-cell-level was 2.79% and on full-cell level 3.06%. By neglecting the DVA on full-cell level for the above reasons, the mass fraction of silicon can be determined to an average of 2.43%. In comparison with the simulation results by Ai et al. (2022) for the LG M50T, the silicon volume fraction in the negative electrode was fitted via simulation to 2% in their study,¹⁶ which is consistent with the findings of the present work. Boveleth et al. (2024) obtained the silicon volume fraction to 2.26% of the total active material fraction by using focused ion beam (FIB)/SEM.⁸⁰ All reported findings from the literature regarding the silicon fraction of the M50T are consistent with the results obtained in this study.

Galvanostatic intermittent titration technique.—The GITT measurement was conducted on half-cell level as outlined in the methodology to determine the diffusion coefficients D_s for the negative and positive electrodes of the M50T and M50LT separately. After analyzing each current pulse as illustrated in Fig. 3 according to Ref. 34 and, 35 the calculated diffusion coefficients were plotted as a function of the lithiation state which can be seen in Fig. 15. The appropriate potential curves are presented in the supplementary information in Fig. S13.

The diffusion coefficients of the negative electrode of the M50T and M50LT (Fig. 15a) generally show a similar course. The initial drop of the diffusion coefficient for low relative capacities occurs due to the interaction with silicon oxide particles. In addition, the local minimum at low relative capacities is shifted toward higher capacities because silicon is more active in that region. The diffusion coefficients also fluctuate more for the M50T compared to the M50LT. The M50T has lower minimum diffusion coefficients but the maximums are similar to that of the M50LT. The M50 analyzed by Chen et al. (2020)¹⁵ shows a similar behavior as the M50T due to their related chemical electrode compositions. Similarly to the inflection points shown in the DVA, the minima and maxima in the diffusion coefficients over the relative capacity can be attributed to phase changes in the active materials which lead to changes in the diffusion coefficients.

For the positive electrode, the diffusion coefficients are more constant over the relative capacity. At high relative capacities, the diffusion coefficients are the lowest. Comparing the M50T with M50LT, it can be seen that the course is similar but the coefficients are generally lower for the M50LT. This could be due to the higher nickel content present in the M50LT and because of the smaller mean particle size. The diffusion coefficients for lithiation and delithiation do only vary slightly.

The diffusion coefficients are then averaged over the relative capacities and summarized in Table I. The diffusion coefficients for the M50T are the highest and in the same order of magnitude as determined by Chen et al.¹⁵ for the M50. The M50LT has the lowest average diffusion coefficients which are about one order of magnitude smaller than those of the other two cells. On the negative electrode side this could result from the absence of silicon; on the positive electrode side the higher nickel content could be responsible for this phenomena.

To confirm the current pulse duration by the dimensionless time scale $\hat{\tau}$ specified in (11), the maximum diffusion coefficients determined in the experiment and the minimum particle radii are considered which leads to a maximum dimensionless time scale $\hat{\tau}$. For the positive electrode of the M50T, the maximum diffusion coefficient is $D_{s,\text{max}} = 6.55 \cdot 10^{-15} \text{ m}^2 \text{ s}^{-1}$ and the minimum mean particle radius is $4.27 \text{ } \mu\text{m}$. With a pulse duration of $\tau = 36 \text{ s}$, the dimensionless timescale is $\hat{\tau} = 0.013 < 0.02$ and thus the criteria for the suitable pulse duration is confirmed. Analogously, the dimensionless time scale $\hat{\tau}$ is calculated for the negative and positive electrode of the M50LT to $\hat{\tau} = 0.0042 < 0.02$ and $\hat{\tau} = 0.0031 < 0.02$, respectively. For the negative electrode of the M50T containing small silicon oxide particles, the volume expansion of the silicon oxide particles needs to be considered. According to, Ref.⁵ the volume expansion is approximately 118% which

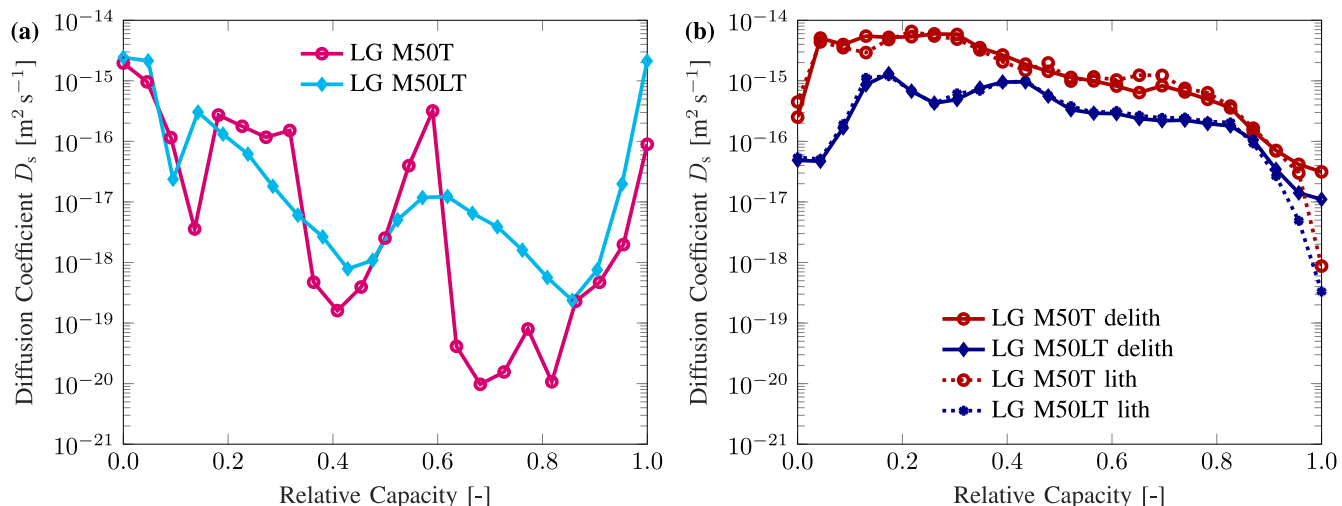


Figure 15. Galvanostatic intermittent titration technique used to determine the diffusion coefficients D_s for the M50T and M50LT for the negative (a) and positive electrodes (b). The latter is separated for lithiation and delithiation.

increases the particle radius by about 29.7% from the fully delithiated and lithiated state. Therefore, the smallest radius measured at almost fully delithiated state is $1.89 \mu\text{m}$. for the fully lithiated state, the radius increases to $2.3 \mu\text{m}$. Since silicon is mainly active below 20% of relative capacity (as discussed in the previous sections), the maximum diffusion coefficient needs to be chosen in that region. The dimensionless time scale was calculated for the different particle radii and diffusion coefficients for silicon oxide below a relative capacity of the composite electrode of 20%. The maximum dimensionless time scale was found for the first current pulse to $\hat{\tau} = 0.0199 < 0.02$. Consequently, an appropriate current pulse of $\tau = 36 \text{ s}$ was chosen in the GITT experiments conducted in this study for all of the electrodes.

By applying (13), the ionic conductivity σ_i can be determined. The ionic conductivity for the M50T at a full-cell voltage of 2.5 V was found to be 1.3 mS m^{-1} and 9.0 mS m^{-1} for the negative and positive electrode, respectively. For the negative electrode of the M50LT, an ionic conductivity of 0.429 mS m^{-1} was determined; for

the positive electrode of the M50LT, σ_i was calculated to 9.2 mS m^{-1} for a full-cell voltage of 2.5 V.

Comparing the calculated ionic conductivity σ_i with the measured total conductivity determined with the four-probe method, it can be seen that the total conductivity is at least three magnitudes higher than the ionic conductivity for all electrodes at a full-cell voltage of 2.5 V. Hence, the total conductivity measured with the four-probe method can practically be interpreted as the electronic conductivity. This confirms the assumption that the ionic conductivity is the limiting conductivity and not the electronic conductivity of the chemical diffusion processes in the electrodes—at least for the full-cell voltage of 2.5 V. Therefore, (13) can be used.^{34,35}

Electrochemical impedance spectroscopy.—EIS data was recorded between the current pulses of the GITT measurement at 25 °C and with varying capacity. In Fig. 16, the EIS data is illustrated for five different relative capacities as examples for the negative and positive electrodes of the M50T and M50LT.

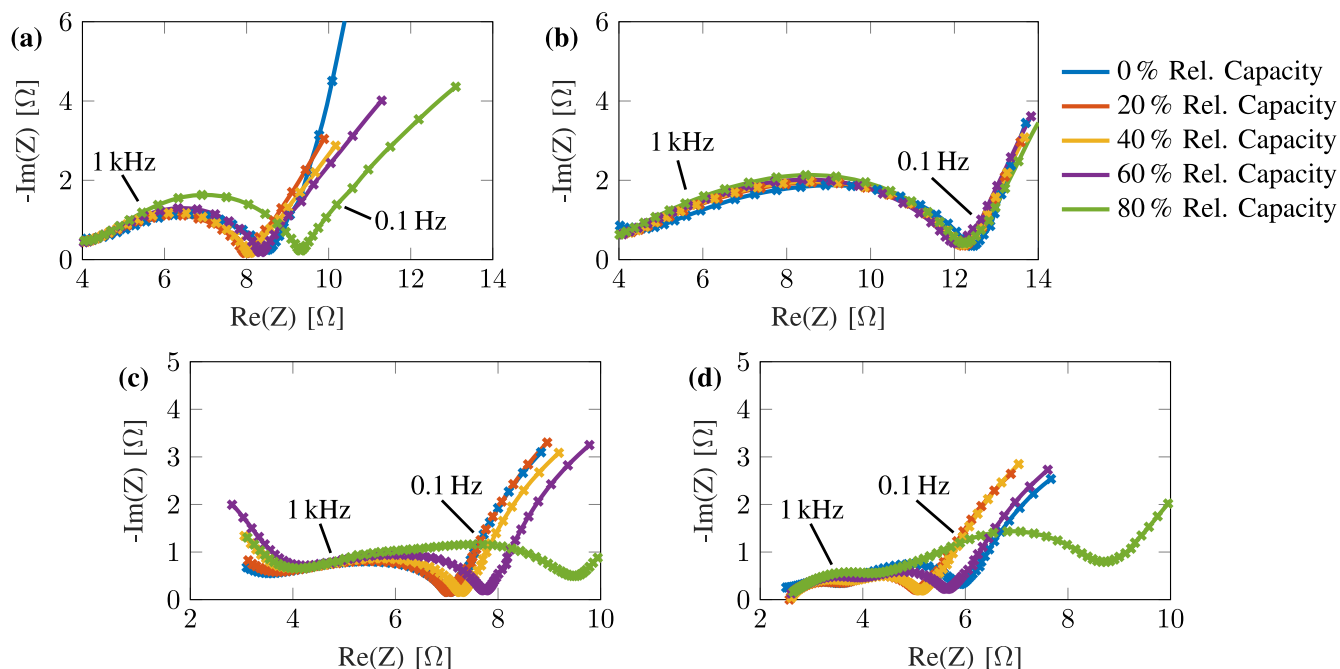


Figure 16. Electrochemical impedance spectroscopy for the negative electrodes of the M50T (a) and M50LT (b) and for the positive electrodes of the M50T (c) and M50LT (d) for relative capacities of 0%, 20%, 40%, 60%, and 80%.

The EIS data for the negative electrodes show one semi-cycle. The negative electrode of the M50LT has generally higher resistances which are also less frequency-sensitive compared to the M50T. For the positive electrodes two smaller semi-cycles can be observed. The first one occurs at high frequencies and the second one at lower frequencies. Both semi-cycles become large with increasing relative capacity. The first semi-cycle for the M50T is less distinctive compared to that of the M50LT.

The results of the EIS measurements can be used to fit equivalent circuit models for the M50T and M50LT and to determine the exchange current density to parameterize electrochemical p2D models.

Conclusion and Future Work

This study analyzed and compared the physical, chemical and electrochemical properties of three commercially available cylindrical battery cells from LG Chem in the 21700 format (LG M50, LG M50T and LG M50LT). Investigations on cell, electrode and particle level were performed. For this, the cylindrical battery cells have been disassembled and the electrode material was extracted. The geometrical dimensions of the electrodes were examined in the physical analysis as well as the electronic conductivity and the X-ray diffractograms. SEM pictures provided information about the particle size distributions and the particle shapes. Further, EDS and GDOES were used to determine the elemental composition of the electrodes on a microscopic and macroscopic scale, respectively. For examining the electrochemical properties, experimental cells in three-electrode configurations including a lithium reference ring were built by using the extracted electrode material of the cylindrical cells. The experimental cells were taken to determine the pOCP of the single electrodes, to calculate the diffusion coefficient by applying GITT and to perform EIS measurements on electrode level. The electrochemical results were post-processed to evaluate the silicon content in the negative composite electrodes, to determine the electrode balancing in the pristine full-cells and to investigate phase changes during (de-)lithiation in DVAs. Different techniques were applied to determine the silicon fraction in the composite negative electrodes: EDS, GDOES, simulated composite OCP, half-cell and full-cell DVA. The average silicon mass fraction in the negative electrode of the M50T was calculated to 2.43% which aligns with results from literature.

The major differences between the M50, M50T and M50LT discovered in this publication are the following: First, it was found out that the M50 and M50T consist of a graphite/silicon oxide composite negative electrode whereas the negative electrode of the M50LT, the latest of the three subvariants, is only made of graphite as an active material. Therefore, the physical, chemical and electrochemical properties of the negative electrode of the M50LT differ compared to those of the other two batteries. The specific capacities of the silicon-containing electrodes were shown to be higher compared to the pure graphite one. Second, the positive electrode of the M50LT has a higher nickel content and thus a higher specific capacity compared to the M50 and M50T. However, all of them can be classified as NMC-811.

Despite the presence of silicon in the negative electrodes, the M50 and M50T could not achieve significantly higher capacities on cylindrical cell level compared to the M50LT (only 1.6% less capacity specified for the M50LT). Due to the higher specific capacity attributed to the higher nickel content in the positive electrode of the M50LT, the positive electrode could be designed smaller by keeping the storage capacity nearly constant. The additional space in the cylindrical cell created thereby could be filled by the coating volume of the negative electrode. In other words, the reduction in capacity due to the omission of silicon was compensated by increasing the negative electrode volume of the M50LT. The space required for that was delivered by a smaller positive electrode with a higher specific capacity. To not limit the

diffusion process by increasing only the electrodes' thickness, the electrodes of the M50LT were designed longer but thinner.

This raised the question if silicon is not needed for high-energy cells anymore and if it is therefore replaceable—at least for the manufacturer LG Chem. Due to its high specific capacity, silicon and silicon oxide are generally considered as promising candidates for increasing the energy density of lithium-ion batteries. Nevertheless, their high volume changes during (de-)lithiation lead to accelerated degradation. Without silicon, the cycle life could be enhanced but the energy density is theoretically lower at the begin of the lifetime. The three battery cells investigated in this study have a specific capacity of roughly 73 mAh g⁻¹ on cylindrical cell level. Especially for mobile applications such as aviation, aerospace or vehicles, higher energy densities are essential for fulfilling the requirements of light-weight transportation. Higher capacities on cell level can mainly be achieved by improving the active materials—for instance by adding energy-dense silicon to the negative electrodes. As a perspective, by combining the nickel-rich positive electrode of the M50LT with the silicon-containing negative electrode of the M50T, the energy density could be increased significantly on a 21700 cylindrical cell level.

To answer the question if silicon is replaceable shortly: Silicon is not replaceable if high energy densities are required. Thus, it can still be seen as an attractive candidate for future lithium-ion battery systems due to its high specific capacity provided that the aging behavior will be improved.

Therefore, silicon's rapid degradation needs to be further investigated to combine both cycle life and high specific capacities. This can eventually be achieved by handling the operating conditions of silicon-containing electrodes appropriately. Associated with this, further investigations in terms of the mechanical behavior such as particle cracking of the silicon-containing cells need to be done to produce high-energy and also long-life lithium-ion batteries. In addition, especially in the aerospace sector, more powerful but more expensive on-board diagnostic systems can be utilized to monitor and improve the electrodes' aging in-operando.

Although the batteries' name implies that the investigated cells could be identical, the analysis conducted here revealed the differences in their properties. Consequently, the batteries need to be characterized appropriately to produce valid simulation data and to interpret experimental results correctly. The results outlined in this study can be used to determine relevant electrode properties for parametrizing battery simulation models for the LG M50T and LG M50LT, e. g. for p2D models based on the approach from Doyle, Fuller and Newman.²¹ For instance, the lithium concentration in the electrodes can be determined with the provided electrode specific capacity and the active material fraction. By using GITT and EIS data, the reaction rate and exchange current densities can be determined.

Acknowledgments

This work is part of the project VisEl (grant number 03XP0529E) funded by the Federal Ministry of Education and Research, Germany (BMBF).

We would like to thank Carola Gaßmann for her experimental support in the laboratory. Thanks to Johann Schimon for the graphical editing of Fig. 2 and for taking the photos of the cell teardown shown in the supplementary data. Icons were taken from Flaticon.

CRediT Authorship Contribution Statement

Nicolas Stapf: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data Curation, Visualization, Writing - Original Draft. Nicolas Schneider: Methodology, Writing - Review & Editing. Meichen Zhan: Investigation, Writing - Review & Editing. Christopher Mehlich: Investigation, Writing - Review & Editing. Kai Peter Birke: Writing - Review & Editing, Supervision, Funding acquisition. Alexander Fill: Writing - Review & Editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The data gathered in the experimental work of this study, supporting the findings of this work, are available from the corresponding author upon reasonable request.

ORCID

Nicolas Stapf  <https://orcid.org/0009-0000-1832-4546>

Nicolas Schneider  <https://orcid.org/0009-0006-6190-4830>

Meichen Zhan  <https://orcid.org/0009-0007-3314-2768>

Christopher Mehlich  <https://orcid.org/0009-0003-6984-7349>

Kai Peter Birke  <https://orcid.org/0000-0002-9679-2369>

Alexander Fill  <https://orcid.org/0000-0002-2414-8750>

References

1. M. N. Obrovac and V. L. Chevrier, "Alloy negative electrodes for Li-ion batteries." *Chem. Rev.*, **114**, 11444 (2014).
2. W. Tao, P. Wang, Y. You, K. Park, C.-Y. Wang, Y.-K. Li, F.-F. Cao, and S. Xin, "Strategies for improving the storage performance of silicon-based anodes in lithium-ion batteries." *Nano Research*, **12**, 1739 (2019).
3. Y. Lu, L. Yu, and X. W. Lou, "Nanostructured conversion-type anode materials for advanced lithium-ion batteries." *Chem.*, **4**, 972 (2018).
4. X. Y. Song, K. Kinoshita, and T. D. Tran, "Microstructural characterization of lithiated graphite." *J. Electrochem. Soc.*, **143**, L120 (1996).
5. H. Pegel, O. von Kessel, P. Heugel, T. Deich, J. Tübke, K. Birke, and D. U. Sauer, "Volume and thickness change of NMC811/SiO₂-graphite large-format lithium-ion cells: from pouch cell to active material level." *Journal of Power Sources*, **537**, 231443 (2022).
6. M. P. Bonkile, Y. Jiang, N. Kirkaldy, V. Sulzer, R. Timms, H. Wang, G. Offer, and B. Wu, "Coupled electrochemical-thermal-mechanical stress modelling in composite silicon/graphite lithium-ion battery electrodes." *Journal of Energy Storage*, **73**, 108609 (2023).
7. X. Zhang, Q. Wang, Y. Li, G. Sun, X. Yu, and H. Li, "Crack-induced abrupt capacity degradation in commercial LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂ (NMC811)/SiO₂-graphite pouch batteries." *RSC Adv.*, **14**, 19116 (2024).
8. J. S. Edge et al., "Lithium ion battery degradation: what you need to know." *Physical Chemistry Chemical Physics : PCCP*, **23**, 8200 (2021).
9. J.-H. Kim, C.-M. Park, H. Kim, Y.-J. Kim, and H.-J. Sohn, "Electrochemical behavior of sio anode for Li secondary batteries." *Journal of Electroanalytical Chemistry*, **661**, 245 (2011).
10. M. Ashuri, Q. He, and L. L. Shaw, "Silicon oxides for Li-ion battery anode applications: Toward long-term cycling stability." *Journal of Power Sources*, **559**, 232660 (2023).
11. T. M. M. Heenan, A. Inawali, M. D. R. Kok, T. G. Tranter, C. Tan, A. Dimitrijevic, R. Jervis, D. J. L. Brett, and P. R. Shearing, "An advanced microstructural and electrochemical datasheet on 18650 Li-ion batteries with nickel-rich NMC811 cathodes and graphite-silicon anodes." *J. Electrochem. Soc.*, **167**, 140530 (2020).
12. P. Heugel, W. Märkle, T. Deich, O. von Kessel, and J. Tübke, "Thickness change and jelly roll deformation and its impact on the aging and lifetime of commercial 18650 cylindrical Li-ion cells with silicon containing anodes and nickel-rich cathodes." *Journal of Energy Storage*, **53**, 105101 (2022).
13. A. Fill, N. van Arnhem, M. Schuhmann, S. Seemann, and K. P. Birke, "An experimental- and simulation-based performance evaluation of a commercial high-energy cell - the path towards electric aviation." *IEEE International Conference on Electrical Systems for Aircraft, Railway, Ship Propulsion and Road Vehicles & International Transportation Electrification Conference (ESARS-ITEC)*, **2024**, 1–8 (2024).
14. A. Aufschläger, A. Durdal, and A. Jossen, "Identifying the origin of inhomogeneous degradation and irreversible swelling inside cylindrical 21700 high-energy lithium-ion cells." *Journal of Energy Storage*, **90**, 111726 (2024).
15. C.-H. Chen, F. Brosa Planella, K. O'Regan, D. Gastol, W. D. Widanage, and E. Kendrick, "Development of experimental techniques for parameterization of multi-scale lithium-ion battery models." *J. Electrochem. Soc.*, **167**, 080534 (2020).
16. W. Ai, N. Kirkaldy, Y. Jiang, G. Offer, H. Wang, and B. Wu, "A composite electrode model for lithium-ion batteries with silicon/graphite negative electrodes." *Journal of Power Sources*, **527**, 231142 (2022).
17. K. O'Regan, F. Brosa Planella, W. D. Widanage, and E. Kendrick, "Thermal-electrochemical parameters of a high energy lithium-ion cylindrical battery." *Electrochimica Acta*, **425**, 140700 (2022).
18. S. E. J. O'Kane, W. Ai, G. Madabattula, D. Alonso-Alvarez, R. Timms, V. Sulzer, J. S. Edge, B. Wu, G. J. Offer, and M. Marinescu, "Lithium-ion battery degradation: how to model it." *Physical chemistry chemical physics : PCCP*, **24**, 7909 (2022).
19. M. P. Bonkile, Y. Jiang, N. Kirkaldy, V. Sulzer, R. Timms, H. Wang, G. Offer, and B. Wu, "Is silicon worth it? modelling degradation in composite silicon—graphite lithium-ion battery electrodes." *Journal of Power Sources*, **606**, 234256 (2024).
20. T. L. Kirk, C. P. Please, and S. J. Chapman, "Physical modelling of the slow voltage relaxation phenomenon in lithium-ion batteries." *J. Electrochem. Soc.*, **168**, 060554 (2021).
21. M. Doyle, T. F. Fuller, and J. Newman, "Modeling of galvanostatic charge and discharge of the lithium/polymer/insertion cell." *J. Electrochem. Soc.*, **140**, 1526 (1993).
22. N. Kirkaldy, M. A. Samieian, G. J. Offer, M. Marinescu, and Y. Patel, "Lithium-ion battery degradation: measuring rapid loss of active silicon in silicon-graphite composite electrodes." *ACS Appl. Energy Mater.*, **5**, 13367 (2022).
23. N. Kirkaldy, M. A. Samieian, G. J. Offer, M. Marinescu, and Y. Patel, "Lithium-ion battery degradation: Comprehensive cycle ageing data and analysis for commercial 21700 cells." *Journal of Power Sources*, **603**, 234185 (2024).
24. S. Baazouzi, N. Feistel, J. Wanner, I. Landwehr, A. Fill, and K. P. Birke, "Design, properties, and manufacturing of cylindrical li-ion battery cells—a generic overview." *Batteries*, **9**, 309 (2023).
25. M. Galasso, S. Schwunk, and V. Knap, "Variations in open circuit voltage of cycled lithium-ion batteries due to silicon-enhanced anodes." *International Spring Seminar on Electronics Technology (ISSE)*, **2024 47th**, 1–55/15/2024 - 5/19/2024. IEEEPrague, Czech Republic.
26. M. Lecompte, M. Bardi, L. Richardet, S. Chevillard, S. Abada, H. Khaled, and S. de Persis, "Experimental characterization of the variability of the thermal runaway phenomenon of a li-ion battery." *SAE Int. J. Adv. & Curr. Prac. in Mobility*, **6**(4), 1777 (2024).
27. L. Richardet, S. de Persis, M. Bardi, L. M., and B. V., "Thermal runaway characterization in an optically accessible vessel: effect of battery cell chemistry and state of charge." *SAE Technical Paper Series*, ed. L. Richardet (2023), SAE Technical Paper Series. SAE International400 Commonwealth Drive, Warrendale, PA, United States.
28. J. Knox, M. Blyth, and A. Hales, "Advancing state estimation for lithium-ion batteries with hysteresis through systematic extended kalman filter tuning." *Sci. Rep.*, **14**, 1 (2024).
29. F. C. Krause, J. P. Ruiz, S. C. Jones, E. J. Brandon, E. C. Darcy, C. J. Iannello, and R. V. Bugga, "Performance of commercial li-ion cells for future nasa missions and aerospace applications." *J. Electrochem. Soc.*, **168**, 040504 (2021).
30. M. Flügel, K. Richter, M. Wohlfahrt-Mehrens, and T. Waldmann, "Detection of li deposition on si/graphite anodes from commercial li-ion cells: A post-mortem gdoes depth profiling study." *J. Electrochem. Soc.*, **169**, 050533 (2022).
31. J. Schmitt, M. Schindler, A. Oberbauer, and A. Jossen, "Determination of degradation modes of lithium-ion batteries considering aging-induced changes in the half-cell open-circuit potential curve of silicon—graphite." *Journal of Power Sources*, **532**, 231296 (2022).
32. A. Fill, M. Zhan, N. Stapf, J. Hemmerling, A. Avdyli, J. Petit, P. Heugel, J. Tübke, and K. P. Birke, "Electromechanical Li-ion cell model considering the composite graphite-silicon structure of the negative electrode." *International Conference on Electrical Engineering and Green Energy (CEEGE)*, **2024 7th**, 186 (2024), IEEE, 6/28/ - 7/1/20242024.
33. P. Heugel, *Alterungsverhalten von Li-Ionen-Zellen mit siliziumhaltigen Anoden - Chemische und physikalische Untersuchungen* (Karlsruher Institut für Technologie (KIT), Karlsruhe) (2024), KITopen-ID: 1000167216.
34. S. D. Kang and W. C. Chueh, "Galvanostatic intermittent titration technique reinvented: Part i. a critical review." *J. Electrochem. Soc.*, **168**, 120504 (2021).
35. S. D. Kang, J. J. Kuo, N. Kapate, J. Hong, J. Park, and W. C. Chueh, "Galvanostatic intermittent titration technique reinvented: Part ii. experiments." *J. Electrochem. Soc.*, **168**, 120503 (2021).
36. W. M. Haynes, D. R. Lide, and T. J. Bruno, *CRC Handbook of Chemistry and Physics* (CRC Press, Boca Raton) (2016)978-1-4987-5429-3.
37. J. Shim and K. A. Striebel, "The dependence of natural graphite anode performance on electrode density." *Journal of Power Sources*, **130**, 247 (2004).
38. H.-P. Ma, J.-H. Yang, J.-G. Yang, L.-Y. Zhu, W. Huang, G.-J. Yuan, J.-J. Feng, T.-C. Jen, and H.-L. Lu, "Systematic study of the siox film with different stoichiometry by plasma-enhanced atomic layer deposition and its application in siox/sio2 superlattice." *Nanomaterials (Basel, Switzerland)*, **9** (2019).
39. J. Landesfeind, J. Hattendorff, A. Ehrl, W. A. Wall, and H. A. Gasteiger, "Tortuosity determination of battery electrodes and separators by impedance spectroscopy." *J. Electrochem. Soc.*, **163**, A1373 (2016).
40. N. Schneider, N. Stapf, A. Bismas, C. Mehlich, K. Peter Birke, and A. Fill, "Parametrization and validation of a doyle-fuller-newman model for commercial sodium-ion batteries: Part I. parametrization." (2025), Unpublished Manuscript.
41. I. Miccoli, F. Edler, H. Pfnür, and C. Tegenkamp, "The 100th anniversary of the four-point probe technique: the role of probe geometries in isotropic and anisotropic systems." *Journal of Physics: Condensed Matter: an Institute of Physics Journal*, **27**, 223201 (2015).
42. J. Sturm, A. Rheinfeld, I. Zilberman, F. B. Spingler, S. Kosch, F. Frie, and A. Jossen, "Modeling and simulation of inhomogeneities in a 18650 nickel-rich, silicon-graphite lithium-ion cell during fast charging." *Journal of Power Sources*, **412**, 204 (2019).
43. M. Heß and P. Novák, "Shrinking annuli mechanism and stage-dependent rate capability of thin-layer graphite electrodes for lithium-ion batteries." *Electrochimica Acta*, **106**, 149 (2013).
44. H. M. Dahn, A. J. Smith, J. C. Burns, D. A. Stevens, and J. R. Dahn, "User-friendly differential voltage analysis freeware for the analysis of degradation mechanisms in li-ion batteries." *J. Electrochem. Soc.*, **159**, A1405 (2012).
45. M. Verbrugge, D. Baker, and X. Xiao, "Formulation for the treatment of multiple electrochemical reactions and associated speciation for the lithium-silicon electrode." *J. Electrochem. Soc.*, **163**, A262 (2016).
46. T. Schied, A. Nickol, C. Heubner, M. Schneider, A. Michaelis, M. Bobeth, and G. Cuniberti, "Determining the diffusion coefficient of lithium insertion cathodes from gitt measurements: Theoretical analysis for low temperatures*." (2021).

- Chemphyschem: A European Journal of Chemical Physics and Physical Chemistry*, **22**, 885 (2021).
47. J. Günther, D. Wycisk, R. Siva Dev, A. Fill, K. Peter Birke, R. Moos, J. Philipp Schmidt, and M. Oldenburger, "Determination of the solid state diffusion coefficient of Li-ion battery single-phase materials using thin model electrodes." *J. Electrochem. Soc.*, **172**, 030525 (2025).
 48. C. Pan, A. P. S. Gaur, M. Lynn, M. P. Olson, G. Ouyang, and J. Cui, "Enhanced electrical conductivity in graphene—copper multilayer composite." *AIP Adv.*, **12** (2022).
 49. F. V. Grechnikov, E. G. Dem'yanenko, and I. P. Popov, "Development of the manufacturing process of aluminum alloys with high strength and conductance." *Russian Journal of Non-Ferrous Metals*, **56**, 15 (2015).
 50. S. Qiao, X. Meng, W. Cao, S. Yu, C. Liu, and Q. Huang, "Effect of lithium salts LiPF₆ and LiBF₄ on combustion properties of electrolyte with EC/PC/EMC under different pressures." *Case Studies in Thermal Engineering*, **30**, 101741 (2022).
 51. R. Weber, C. R. Fell, J. R. Dahn, and S. Hy, "Operando X-ray diffraction study of polycrystalline and single-crystal Li_xNi_{0.5}Mn_{0.5}Co_{0.5}O₂." *J. Electrochem. Soc.*, **164**, A2992 (2017).
 52. M. Akhilash, P. S. Salini, B. John, N. Supriya, S. Sujatha, and T. D. Mercy, "Thermal stability as well as electrochemical performance of Li-rich and Ni-rich cathode materials—a comparative study." *Ionics*, **29**, 983 (2023).
 53. J. Li, Y. Li, W. Yi, and P. Ma, "Improved electrochemical performance of cathode material LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂ by doping magnesium via co-precipitation method." *J. Mater. Sci.: Mater. Electron.*, **30**, 7490 (2019).
 54. J. Li and J. R. Dahn, "An in situ X-ray diffraction study of the reaction of Li with crystalline Si." *J. Electrochem. Soc.*, **154**, A156 (2007).
 55. J. Park, S. S. Park, and Y. S. Won, "In situ xrd study of the structural changes of graphite anodes mixed with SiO_x during lithium insertion and extraction in lithium ion batteries." *Electrochimica Acta*, **107**, 467 (2013).
 56. A. Kazzazi, D. Bresser, A. Birrozzi, J. von Zamory, M. Hekmatfar, and S. Passerini, "Comparative analysis of aqueous binders for high-energy li-rich nmc as a lithium-ion cathode and the impact of adding phosphoric acid." *ACS Applied Materials & Interfaces*, **10**, 17214 (2018).
 57. X.-G. Sun, C. J. Jafra, S. Tan, A. Borisevich, R. B. Gupta, and M. Paranthaman, "Facile surface coatings for performance improvement of NMC811 battery cathode material." *J. Electrochem. Soc.*, **169**, 020565 (2022).
 58. J.-J. Adjizian, C. D. Latham, S. Öberg, P. R. Briddon, and M. I. Heggie, "Dft study of the chemistry of sulfur in graphite, including interactions with defects, edges and folds." *Carbon*, **62**, 256 (2013).
 59. P. Verma, P. Maire, and P. Nov, "A review of the features and analyses of the solid electrolyte interphase in Li-ion batteries." *Electrochimica Acta*, **55**, 6332 (2010).
 60. S. K. Heiskanen, J. Kim, and B. L. Lucht, "Generation and evolution of the solid electrolyte interphase of lithium-ion batteries." *Joule*, **3**, 2322 (2019).
 61. S. J. An, J. Li, C. Daniel, D. Mohanty, S. Nagpure, and D. L. Wood, "The state of understanding of the lithium-ion-battery graphite solid electrolyte interphase (SEI) and its relationship to formation cycling." *Carbon*, **105**, 52 (2016).
 62. Z. Shadike, Y. Chen, E. Hu, J. Zhang, and X.-Q. Yang, "Interphasial engineering for Ni-rich nmc cathode materials." *Trends in Chemistry*, **5**, 775 (2023).
 63. L. Xiaobing, J. Holland, S. Burgess, S. Bhadare, S. Yamaguchi, D. Birtwistle, P. Statham, and N. Rowlands, "Detection of lithium X-rays by eds." *Microscopy and Microanalysis*, **19**, 1136 (2013).
 64. J. Kasnatscheew, S. Röser, M. Börner, and M. Winter, "Do increased ni contents in LiNi_xMn_yCo_zO₂ (NMC) electrodes decrease structural and thermal stability of Li ion batteries? A thorough look by consideration of the Li+ extraction ratio." *ACS Appl. Energy Mater.*, **2**, 7733 (2019).
 65. A. Gomez Martin, F. Reissig, L. Frankenstein, M. Heidbüchel, M. Winter, T. Placke, and R. Schmich, "Magnesium substitution in Ni-rich NMC layered cathodes for high-energy lithium ion batteries." *Adv. Energy Mater.*, **12** (2022).
 66. M. Li and J. Lu, "Cobalt in lithium-ion batteries." *Science (New York, N.Y.)*, **367**, 979 (2020).
 67. W. Li, S. Lee, and A. Manthiram, "High-nickel NMA: A cobalt-free alternative to NMC and NCA cathodes for lithium-ion batteries." *Advanced materials (Deerfield Beach, Fla.)*, **32**, e2002718 (2020).
 68. Z. Shadike, Y. Chen, L. Liu, X. Cai, S. Shen, and J. Zhang, "Improved cyclic stability of LiNi_{0.8}Mn_{0.1}Co_{0.1}O₂ cathode enabled by a novel cei forming additive." *Frontiers in Energy*, **18**, 535 (2024).
 69. S.-B. Son, D. Robertson, Y. Tsai, S. Trask, A. Dunlop, and I. Bloom, "Systematic study of the cathode compositional dependency of cross-talk behavior in Li-ion battery." *J. Electrochem. Soc.*, **167**, 160508 (2020).
 70. M. Ge, C. Cao, G. M. Biesold, C. D. Sewell, S.-M. Hao, J. Huang, W. Zhang, Y. Lai, and Z. Lin, "Recent advances in silicon-based electrodes: from fundamental research toward practical applications." *Advanced materials (Deerfield Beach, Fla.)*, **33**, e2004577 (2021).
 71. T. Waldmann, Björn-Ingo Hogg, and M. Wohlfahrt-Mehrens, "Li plating as unwanted side reaction in commercial Li-ion cells—a review." *Journal of Power Sources*, **384**, 107 (2018).
 72. J. W. Wang et al., "Two-phase electrochemical lithiation in amorphous silicon." *Nano Lett.*, **13**, 709 (2013).
 73. Z. Liu, Q. Yu, Y. Zhao, R. He, M. Xu, S. Feng, S. Li, L. Zhou, and L. Mai, "Silicon oxides: a promising family of anode materials for lithium-ion batteries." *Chem. Soc. Rev.*, **48**, 285 (2019).
 74. T. Chen, J. Wu, Q. Zhang, and X. Su, "Recent advancement of SiO_x based anodes for lithium-ion batteries." *Journal of Power Sources*, **363**, 126 (2017).
 75. K. Märker, P. J. Reeves, C. Xu, K. J. Griffith, and C. P. Grey, "Evolution of structure and lithium dynamics in LiNi_{0.8}Mn_{0.1}Co_{0.1}O₂ (NMC811) cathodes during electrochemical cycling." *Chemistry of Materials*, **31**, 2545 (2019).
 76. R. Jung, M. Metzger, F. Maglia, C. Stinner, and H. A. Gasteiger, "Oxygen release and its effect on the cycling stability of LiNi_xMn_yCo_zO₂ (NMC) cathode materials for Li-ion batteries." *J. Electrochem. Soc.*, **164**, A1361 (2017).
 77. Y. Jiang, G. Offer, J. Jiang, M. Marinescu, and H. Wang, "Voltage hysteresis model for silicon electrodes for lithium ion batteries, including multi-step phase transformations, crystallization and amorphization." *J. Electrochem. Soc.*, **167**, 130533 (2020).
 78. Dahn, "Phase diagram of Li_xC₆." *Physical Review. B, Condensed Matter*, **44**, 9170 (1991), 1991.
 79. H.-J. Noh, S. Youn, C. Seung Yoon, and Y.-K. Sun, "Comparison of the structural and electrochemical properties of layered Li[Ni_xCo_yMn_z]O₂ (x = 1/3, 0.5, 0.6, 0.7, 0.8 and 0.85) cathode material for lithium-ion batteries." *Journal of Power Sources*, **233**, 121 (2013).
 80. L. Boveleth, A. Lindner, W. Menesklo, T. Danner, and A. Latz, "Material parameters affecting li plating in Si/graphite composite electrodes." *Electrochimica Acta*, **506**, 145010 (2024).